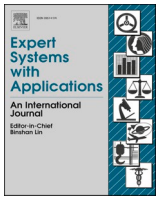
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# Sparrow search algorithm with adaptive *t* distribution for multi-objective low-carbon multimodal transportation planning problem with fuzzy demand and fuzzy time

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ABSTRACT

In this paper, we present a multi-objective low-carbon multimodal transportation planning problem with fuzzy demand and fuzzy time (MOLCMTPP-FDFT) that minimizes both cost and time while incorporating mandatory carbon emission, carbon tax, carbon trading, and carbon offset policies. Chance constrained programming and interval programming are introduced to formulate the fuzzy demand chance constrained programming model and the fuzzy time interval programming model, respectively. Then, the mathematical model of uncertain MOLCMTPP-FDFT is transformed into a deterministic model. Based on the sparrow search algorithm, *t* distribution and the concept of Pareto optimality for multi-objective optimization, we also propose a solution strategy for the proposed model. In this algorithm, the number of iterations is used as the degree of freedom of - to update the sparrow location, which strikes a balance between the capabilities of global search and local search. Finally, the proposed algorithm and MOLCMTPP-FDFT are applied to a real case, resulting in a minimum cost of 260730.48 and a time duration of 13.044, which outperform the minimum cost of 268874.88 and minimum time of 18.32 obtained using single-mode transportation. The carbon emissions resulting from the lowest-cost solution obtained using single-mode transportation are 3,198.48, which are significantly more than the allowed emissions. Therefore, the proposed algorithm and mathematical model of MOLCMTPP-FDFT are valuable tools for optimizing multimodal transportation route. Additionally, the experimental results not only validate the superior efficiency and energy-saving benefits of the proposed multimodal transportation routes in comparison to the actual single-modal transportation, but also demonstrate the applicability of different low-carbon policies.

## 1. Introduction

With the rapid development of logistics, the implementation of transportation structure adjustment strategy has been vigorously promoted in recent years with the aim of facilitating low-carbon transportation, reducing logistics costs and enhancing transportation efficiency (Li, Liu, & Yang, 2021). Multimodal transportation is a crucial component of transportation infrastructure optimization and has emerged as a pivotal logistics mode for numerous enterprises engaged in transport operations (Gurupada, Sankar, & José, 2020). Multimodal transportation refers to the process of transporting cargo to its destination by utilizing two or more modes of transportation in a coordinated manner (Archetti, Peirano, & Speranza, 2022). The optimization of multimodal transportation efficiency has become a central focus of inquiry among numerous scholars (Li et al., 2021; Gurupada, Sankar, & José, 2020; Laurent, Vallerand, Van der Meer, & Sophie, 2020).

The "Belt and Road" Initiative has stimulated growing industry interest in low-carbon multimodal transportation, which plays a critical role as an organizational form within modern logistics and transportation systems (Qi, Harrod, Psaraftis, & Lang, 2022). The transportation sector accounts for up to 14 % of the total global carbon emissions, with road transportation alone contributing to 70 % of these emissions. Therefore, it is crucial to consider carbon emission factors in multimodal transportation processes (Maja, & Alan, 2009). In significant documents, the Chinese government has emphasized that it is necessary to actively adjust the transport structure, promote multimodal transportation, improve the circulation system for green, low-carbon and circular development, and optimize this problem in conjunction

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with various low-carbon policies (Chen, & Wang, 2016). The Low-Carbon Multimodal Transportation Planning Problem (LCMTPP) is the designated term for this issue. Additionally, it takes account into carbon emission allowances under various low-carbon policies and capacity limitations between road segments and transit nodes (Archetti, Peirano, & Speranza, 2022). In this paper, multiple modes of transportation, including trucks, trains and ships, are used to transport goods from the point of origin through multiple cities to the final destination. To solve the LCMTPP for a given set of origin–destination transport goods, it is necessary to effectively integrate multiple transport modes and tools, optimize transit nodes and routes, and minimize carbon emissions.

The competition among logistics enterprises is gradually intensifying as transportation means continue to improve, and multimodal transportation networks are progressively enhanced. In order to enhance the core competitiveness of enterprises, it is imperative for them to evaluate the transportation route scheme based on multiple objectives, including minimizing transportation cost and time, reducing carbon emissions, and maximizing customer satisfaction (Adil & Kemal, 2016). Additionally, enterprises should also take into account various uncertain factors in the transportation system such as advance planning of transport schemes, seasonal demand fluctuations, and the influence of human subjectivity on demand (Zhang, Jin, Yuan, & Zhang, 2021). The transportation process may encounter uncontrollable phenomena and problems, such as traffic congestion, equipment failure, or extreme weather conditions. These factors contribute to the unpredictability of transportation time. The decision-makers can effectively design strategies that are capable of adapting to changes in the external environment, while minimizing the impact of uncertainties and ensuring the competitiveness of enterprises. This can be achieved by thoroughly considering various uncertainties in the external environment and utilizing multiple evaluation indicators as references. The promotion of low-carbon transportation is effectively achieved through the assessment and formulation of transportation schemes from multiple perspectives, even under uncertain conditions (Goncalves, Hsu, & Xu, 2009). Therefore, this paper presents a multi-objective low-carbon multimodal transportation planning problem with fuzzy demand and fuzzy time (MOLCMTPP-FDFT), aiming to minimize both transportation cost and time, while integrating various low-carbon policies into the model.

In order to find a feasible solution for balancing transportation cost, carbon emissions, and transportation time under different low-carbon policies. In this paper, a new Pareto-based multi-objective approach is developed, which combines the sparrow search algorithm (SSA) (Xue, & Shen, 2020) and $t$ distribution (Kitagawa, 1953) mutation. The proposed method aims to solve the MOLCMTPP-FDFT and transformation models considering four low-carbon policies, including mandatory carbon emission, carbon tax, carbon trading, and carbon offset. There are some motivations for using the sparrow search algorithm with adaptive $t$ distribution (ATDSSA) to solve this problem which are as follows. Firstly, SSA algorithm is a swarm-based metaheuristic method, which offers several advantages such as minimal parameters, rapid convergence speed, and straightforward implementation. It was primarily utilized for solving deep learning (Fan et al., 2022), machine learning (Jia et al., 2022), feature recognition (Wu, Zhang, Han, Nan, & Li, 2022) and combinatorial optimization problems (Li, & Wang, 2022). It achieved superior solutions in reasonable computing time (Zhang, Zhang, Ma, & Liu, 2019). Secondly, the swarm optimization algorithm maintains a solution pool during a search process (Zhang, Zhang, Ma, & Liu, 2019; Zhang, Liu, Ma, & Zhang, 2020). Thus, swarm optimization algorithms are particularly suitable for finding trade-off solutions in a single run (Zhang, Zhang, Ma, & Liu, 2019; Zhang, Liu, Ma, & Zhang, 2020). Thirdly, metaheuristics are a general algorithmic framework for finding solutions to optimization problems, in which local heuristics are guided and adapted to efficiently explore the solution space (Zhang, Zhang, Ma, & Liu, 2019; Zhang, Liu, Ma, & Zhang, 2020).

The main contributions of this paper are as follows. Firstly, we have introduced and modeled the MOLCMTPP-FDFT, and transformed model under different low-carbon policies. This enables us to evaluate the impact of different low-carbon policies on the multimodal transportation routing scheme. Secondly, in order to obtain high-quality solutions, ATDSSA was designed to solve the MOLCMTPP-FDFT. This approach not only preserves the characteristics of SSA but also utilizes the number of iterations as freedom of $t$ distribution for balancing global and local search abilities, thereby preventing premature convergence or falling into local optima. Thirdly, chance constrained programming and interval programming are introduced to address uncertain demand and time. The multi-objective fuzzy chance constrained programming model and interval programming model are constructed based on the actual situation. Furthermore, the proposed ATDSSA incorporates the fast non-dominated sorting method along with its underlying concept. Finally, we conduct a comprehensive analysis of the performance of our proposed algorithm. As later reported, our method efficiently generates trade-off solutions for MOLCMTPP-FDFT under different low-carbon policies.

The rest of this paper is organized as follows. Section 2 provides a brief overview of the existing literatures. Section 3 formally describes MOLCMTPP-FDFT and the transformation model under different low-carbon policies. Section 4 introduces several fuzzy approaches, including chance constrained programming based on credibility theory, fuzzy variables, and interval programming. Section 5 presents the sparrow search algorithm with adaptive $t$ distributed based on non-dominated sorting method. Section 6 describes computational experiments that analyzed the performance and parameter setting of ATDSSA, while also applying the proposed MOLCMTPP-FDFT and transformation models under different low-carbon policies in a real case study. Section 7 summarizes the impact of different low-carbon policies on multimodal transportation routing decisions.

## 2. Literature review

The aim of this section is to provide a short but comprehensive overview of work related to low-carbon multimodal transportation planning problems, multi-objective optimization techniques, and fuzzy approaches.

In recent years, the topic of low-carbon in multimodal transportation has attracted significant attention from researchers. Li and Sun (2022) addressed the problem with demand uncertainty and carbon trading price randomness by formulating a robust stochastic optimization model for container multimodal transportation with minimum total cost as the objective function. They also developed a hybrid fireworks algorithm with gravitational search operator (FAGSO) to solve it, which demonstrated its effectiveness. (Li, & Sun, 2022). Cheng (2022) established a multimodal transportation planning problem model with the objective of minimizing the total transportation cost, which includes costs of transportation, transit, waiting and carbon emission, respectively (Cheng, 2022). To solve this model effectively, a genetic algorithm incorporating reservation and immigration strategies was employed. This approach provides decision support for selecting an optimal multimodal transportation scheme (Cheng, 2022). Liao, Tseng and Lu (2009) compared the carbon emissions of only truck transportation with the carbon emissions of multimodal transportation. The findings revealed that employing multimodal transportation for freight transportation can significantly reduce carbon dioxide emissions (Liao, Tseng, & Lu, 2009). Behnam, Joseph, Alok and Ali (2015) proposed a strategic supply chain planning model that integrates economic and carbon emission targets within the framework of a carbon tax policy. This nonlinear supply chain planning model is solved using an improved cross-entropy method (Behnam, Joseph, Alok, & Ali, 2015).

There are few studies on low-carbon multimodal transportation planning problems involving multiple objectives in the existing literature. The main reason is that multi-objective problems are more difficult than single-objective ones. Most multimodal transportation planning

problems with multiple objectives are solved by linear weighting into single objective problems. Kalinina, Olsson and Larsson (2013) established an optimization model, which aims to minimize transportation time, cost, and carbon emissions in multimodal transportation planning problem (Kalinina, Olsson & Larsson, 2013). Siu, Lam and Gu (2015) developed innovative methods for designing enhanced intermodal transportation networks provided by freight integrators. They constructed a bi-objective optimization model to minimize both cost and transportation time, and utilized CPLEX, which is a commonly used solver for mixed integer linear programming (MILP) problems, to find Pareto solutions (Siu, Lam, & Gu, 2015). With the increase in transit nodes, the combined routes of different transportation modes grow exponentially, resulting in more complex calculations (Xiong & Wang, 2014). Exact algorithms and optimization solvers can only handle small-scale combinatorial optimization problems, making it challenging to provide solutions for large-scale problems within a limited computing time (Zhang, Liu, Ma, & Zhang, 2020). Therefore, intelligent optimization algorithms that can provide approximate optimal solutions to large-scale problems within an ideal time, have widely used by many scholars (Zhang, Liu, Ma, & Zhang, 2020). For instance, genetic algorithm (Xie, & Cao, 2021; Fattahi, & Behnamian, 2021), simulated annealing algorithm (Oudani, 2021), and fireworks algorithm (Mouna, & Sadok, 2017) have demonstrated satisfactory results in solving multi-objective multimodal transportation planning problems.

Besides, scholars have focused on these multimodal transportation planning problems under uncertainty. Considering the uncertain factors into multimodal transportation process can reduce the environmental impacts on the transportation schemes. The multimodal transportation planning problem under uncertainty mainly includes uncertainties in transportation time, volume, and customer demand. Uncertain variables can be classified into random variables and fuzzy variables (Shi, Boudouh, & Grunder, 2017) with different expressions used in routing optimization problems. Stochastic programming is commonly employed to solve problems involving random variables, such as robust optimization (Iyengar, 2005) and auxiliary variable method (Bertsimas, & Sim, 2004). Triangular fuzzy numbers, trapezoidal fuzzy numbers and interval numbers are typically utilized to represent fuzzy variables. Furthermore, expected value models (Xu, Yan, & Li, 2011), chance constrained programming (Messaoud, 2021), and interval programming (Subulan, 2020; Robers, & Ben-Israel, 1969) are applied to transform fuzzy variables into certain ones.

Kalinina, Olsson and Larsson (2013) used a fuzzy chance constraint model to solve the multimodal transportation with uncertain time. Fazayeli, Eydi and Kamalabadi (2018) proposed a mixed integer mathematical fuzzy model for the multimodal transportation planning problem with fuzzy demand. A fuzzy α-cut technology is used to transform the fuzzy demand an expectation interval, and transformed the original fuzzy model into an auxiliary deterministic one. Zhang (2022) considered the optimization of multimodal transportation routing under the double uncertainty in both time window and demand by using trapezoidal fuzzy variables to represent these uncertainties. The fuzzy chance constrained programming was applied to solve the bi-objective fuzzy model. (Zhang, 2022).

The research on MOLCMTPP-FDFT under different low-carbon policies is limited, highlighting the need for extensive investigation in this area. This includes employing hybrid methods that utilize various heuristic evolutionary/swarm algorithms, exact methods, and local search techniques to enhance solution quality and/or reduce computational time (Zhang, Zhang, Ma, & Liu, 2019; Zhang, Liu, Ma, & Zhang, 2020). In this paper, we aim to construct MOLCMTPP-FDFT under various low-carbon policies such as mandatory carbon emission, carbon tax, carbon trading and carbon offset. To address these problems, we propose Pareto-based sparrow search algorithm with adaptive $t$ distributed. Through case analysis, we discussed the influence of different low-carbon policies on multimodal transportation routing decisions. These findings provide valuable insights for decision-makers involved in implementing low-carbon multimodal transportation practices and assist governmental bodies and relevant organizations in formulating future low-carbon policies.

Finally, Table 1 classifies and summarizes literature mentioned above from the aspects of problem categories, uncertainty components of problems, transportation mode, optimization objectives, carbon emissions, low-carbon policies and solution methodology (Subulan, 2020). As shown in Table 1, the literature on multimodal transportation planning problem is abundant. However, there remains a scarcity of studies that consider multiple objectives, various uncertainties, and multiple low-carbon policies. Most of existing literature have concentrated on transportation in deterministic environments or have only taken one uncertain factor into account. They have also neglected low-carbon policies when estimating carbon emissions, which makes them inaccurate representations of reality. When solving the low-carbon multimodal transportation planning problem, various uncertain factors and multiple low-carbon policies need to be considered simultaneously in order to accurately reflect reality.

## 3. Mathematical model for MOLCMTPP-FDFT

The MOLCMTPP-FDFT refers particularly to the problem of effectively transporting a batch of cargo from the origin to the destination via a variety of transportation means, such as trucks, ships, trains, and planes, while taking fuzzy time and demand into account. In this complex transportation system, the effective and rational integration of various transfer nodes and transportation modes is crucial in minimizing the transportation time, cost, and carbon emissions that are converted to costs and associated with low-carbon policies, such as mandatory carbon emission, carbon tax, carbon trading and carbon compensation.

### 3.1. The basic MOLCMTPP-FDFT– model I

Logistics enterprise needs to adopt multiple transportation means to transport some goods with uncertain demand from the origin to the destination through various transit nodes with the minimum transportation cost and time. In response to national initiatives promoting low-carbon transportation, these enterprises must develop corresponding schemes for low-carbon transportation under different low-carbon policies. To facilitate model construction, we make the following assumptions: (1) goods cannot be disassembled in transportation process; (2) selecting at most one transportation mode between two transit nodes; (3) each transit node can only carry out mode conversion at most once; (4) capacity constraint should be considered for each transportation section and transit node; (5) external factors such as weather conditions, equipment failure or cargo damage are not taken into account.

Sets:

$O$: Point of origin for transportation.

$D$: Transportation destination.

$E$: Set of nodes in a multimodal transportation network,.$\{O, D\} \subseteq E$

$E^{-}_{(i)}$: Set of backward nodes of node $i$,.$E^{-}_{(i)} \subseteq E$

$E^{+}_{(i)}$: Set of forward nodes of node $i$,.$E^{+}_{(i)} \subseteq E$

$A$: Set of directed arcs between two transit nodes.

$R$: Set of transportation modes.

$R_{ij}$: Set of transportation modes on a directed arc $(i,j)$,.$R_{ij} \subseteq R$

$R_i$: Set of transportation modes for backward and forward nodes of node $i$,.$R_i \subseteq R$

Parameters:

$d^{r}_{ij}$: Distance from node $i$ to node $j$ using transportation mode.$r$

$\widetilde{q}$: Fuzzy demand.

$c_r$: Unit-distance transportation cost of per unit goods under mode.$r$

$c_{rh}$: Unit cost of transiting transportation from mode $r$ to mode.$h$

$\gamma$: Carbon tax rate under carbon tax policy.

$c_{\text{trading}}$: Unit trading price under carbon trading policy.

**Table 1**
An overview of the articles on multimodal transportation planning problem.

| Article | Problem Type | Uncertainty | Mode | Multi-Objectivity | Carbon Emission | low-carbon Policies Mandatory carbon emission | Carbon tax | Carbon trading | Carbon offset | Solution Methodology |
|---|---|---|---|---|---|---|---|---|---|---|
| (Chen, & Wang, 2016) | Transportation mode selection | Stochastic demand | Multiple | \ | √ | √ | | √ | \ | comparison of the supply and demand risks under three different scenarios & Expected value model |
| (Liao, Tseng, & Lu, 2009) | MTPP | \ | Multiple | | √ | \ | | \ | \ | Activity-based methods |
| (Kalinina, Olsson, & Larsson, 2013) | MTPP | Fuzzy time | Multiple | √ | \ | \ | | \ | \ | Chance constrained programming |
| (Xiong, & Wang, 2014) | MTPP | Time windows | Multiple | √ | \ | | \ | \ | | The bi-level multi-objective Taguchi genetic algorithm |
| (Siu, Lam, & Gu, 2015) | MTPP | \ | Multiple | √ | √ | √ | \ | \ | \ | CPLEX |
| (Mouna, & Sadok, 2017) | MTPP | \ | Multiple | √ | \ | \ | \ | \ | \ | Multi-objective firework algorithm based on the Pareto-dominance method |
| (Gurupada, Sankar, & José, 2020) | MTPP | \ | Multiple | | \ | \ | \ | \ | \ | LINGO & Neural network |
| (Laurent, Vallerand, Van der Meer, & Sophie, 2020) | MTPP | \ | Multiple | √ | √ | \ | | \ | \ | Multicriteria approach & Decision support system |
| (Xie, & Cao, 2021) | MTPP | Interval time & Interval price | Multiple | √ | \ | \ | \ | \ | \ | Genetic algorithm & Interval ranking |
| (Archetti, Peirano, & Speranza, 2022) | MTPP | \ | Multiple | \ | \ | \ | | \ | \ | Literature review & Statistical analysis |
| (Cheng, 2022) | MTPP | \ | Multiple | | √ | \ | \ | \ | \ | The genetic algorithm based on preservation strategy and immigration strategy |
| (Zhang, 2022) | MTPP | Fuzzy demand & Fuzzy time | Multiple | √ | √ | √ | \ | \ | \ | Chance Constrained programming & Adaptive NSGA-II algorithm |
| (Adil, & Kemal, 2016) | MTPP & Load planning problem | Fuzzy goals | Multiple | √ | √ | \ | \ | | \ | MILP & Fuzzy multi-objective optimization & LINGO & compromise programming technique |
| (Fazayeli, Eydi, & Kamalabadi, 2018) | MTPP & LRP | Fuzzy demand & Time windows | Multiple | √ | \ | \ | \ | \ | \ | Genetic algorithm & Expected interval |
| (Fattahi, & Behnamian, 2021) | MTPP & LP | \ | Multiple | √ | \ | \ | \ | \ | | Sub-population genetic algorithm & Elastic method |
| (Oudani, 2021) | MTPP & LP | \ | Multiple | | \ | \ | \ | \ | \ | Simulated annealing algorithm |
| (Li et al., 2021) | CMTPP & Order consolidation | Time windows | Multiple | \ | \ | \ | \ | \ | \ | Kernel search & Dynamic programming & Column generation |
| (Qi, Harrod, Psaraftis, & Lang, 2022) | MTPP & Service selection | Floating carbon cost | Multiple | \ | √ | | \ | | \ | Mixed-integer programming for CMCF & CPLEX |
| (Zhang, Jin, Yuan, & Zhang, 2021) | LCMTPP | Stochastic demand & Random time | Multiple | √ | √ | \ | \ | | \ | Catastrophic adaptive genetic algorithm based on Monte Carlo sampling & Robust stochastic optimization |
| (Li, & Sun, 2022) | LCMTPP | Random demand & Random carbon trading price | Multiple | \ | √ | | \ | √ | | Robust optimization & Fireworks algorithm with gravitational search operator |
| The present paper | LCMTPP | Fuzzy demand & Interval time | Multiple | √ | √ | √ | √ | √ | √ | Chance constrained programming & Interval programming & ATDSSA |

MTPP: Multimodal transportation planning problem; CMTPP: Capacitated multimodal transportation planning problem; LCMTPP: Low-carbon multimodal transportation planning problem; LP: Location problem; MILP: Mixed-integer programming.

$c_{offset}$: Unit offset price under carbon offset policy.

$e_r$: Unit carbon emission at mode.$r$

$e_{rh}$: Unit carbon emission of transiting transportation from mode $r$ to mode.$h$

$\widetilde{t_{ij}^r}$:Actual transportation time from node $i$ to node $j$ using mode.$r$

$[t_{ijr}^L, t_{ijr}^U]$: Fuzzy time interval from node $i$ to node $j$ using mode.$r$

$\widetilde{t_i^{rh}}$: Unit actual transit time when converting from mode $r$ to mode $h$ at node.$i$

$[t_{irh}^L, t_{irh}^U]$: Unit fuzzy transit time interval at node $i$ when converting mode $r$ to mode.$h$

$t_i$: Actual arrival time of node.$i$

$Q_M$: Upper bound of carbon emission under mandatory carbon emission policy.

$Q_T$: Carbon emission allowances under carbon trading policy.

$Q_O$: Carbon emission allowances under carbon offset policy.

$Q_{trading}$: The total amount of carbon trading under carbon trading policy.

$Q_{offset}$: The amount of carbon offset under carbon offset policy.

$Q$: Upper bound of carbon emissions.

$\Omega_{ij}^r$: Maximum transportation capacity for mode $r$ on directed arc.$(i,j)$

$\Psi_i^{rh}$: Maximum transit capacity for converting from mode $r$ to mode $h$ at node.$i$

$n$: The number of nodes in the multimodal network.

Decision variables:

$$X_{ij}^r = \begin{cases} 1, If moderisusedindirectedarc(i,j) \\ 0, \text{otherwise} \end{cases}$$

$$Y_i^{rh} = \begin{cases} 1, Ifconvertingfrommodertohatnodei \\ 0, \text{otherwise} \end{cases}$$

Mathematical formulation

$$\min Z_1 = \sum_{(i,j)\in A}\sum_{r\in R} c_r d_{ij}^r \widetilde{q} X_{ij}^r + \sum_{i\in E\backslash\{O,D\}}\sum_{r\in R}\sum_{h\in R} c_{rh}\widetilde{q} Y_i^{rh} \tag{1}$$

$$\min Z_2 = \sum_{(i,j)\in A}\sum_{r\in R}\left[t_{ijr}^L, t_{ijr}^U\right] X_{ij}^r + \sum_{i\in E\backslash\{O,D\}}\sum_{r\in R}\sum_{h\in R}\left[t_{irh}^L, t_{irh}^U\right]\widetilde{q} Y_i^{rh} \tag{2}$$

s.t

$$\sum_{j\in E_{(i)}^+}\sum_{h\in R_{ij}} X_{ij}^h - \sum_{l\in E_{(i)}^-}\sum_{r\in R_{ij}} X_{li}^r = \begin{cases} 1 & i = O \\ 0 & \forall i \in E\backslash\{O,D\} \\ -1 & i = D \end{cases} \tag{3}$$

$$\sum_{r\in R_{ij}} X_{ij}^h \le 1 \,\forall (i,j) \in A \tag{4}$$

$$\sum_{r\in R_i}\sum_{h\in R_i} Y_i^{rh} \le 1 \quad \forall i \in E\backslash\{O,D\} \tag{5}$$

$$\sum_{l\in E_{(i)}^-} X_{li}^r - \sum_{r\in R_i} Y_i^{rh} = 0 \,\forall i \in E\backslash\{O,D\}, \forall h \in R_i \tag{6}$$

$$\sum_{r\in R_i} Y_i^{rh} - \sum_{j\in E_{(i)}^+} X_{ij}^h = 0 \,\forall i \in E\backslash\{O,D\}, \forall h \in R_i \tag{7}$$

$$Q_{\text{emission}} = \sum_{(i,j)\in A}\sum_{r\in R} \widetilde{q} e_r d_{ij}^r X_{ij}^r + \sum_{i\in E}\sum_{r\in R}\sum_{h\in R} \widetilde{q} e_{rh} Y_i^{rh} \tag{8}$$

$$\Omega_{ij}^r\left(X_{ij}^r - 1\right) \le \widetilde{q} \le \Omega_{ij}^r X_{ij}^r \,\forall (i,j) \in A, \forall r \in R_{ij} \tag{9}$$

$$\Psi_i^{rh}\left(Y_i^{rh} - 1\right) \le \widetilde{q} \le \Psi_i^{rh} Y_i^{rh} \,\forall i \in E\backslash\{O,D\}, \forall r,h \in R_i \tag{10}$$

$$t_j = t_i + \sum_{h\in R_{ij}} t_{ij}^h X_{ij}^h + \sum_{h\in R}\sum_{r\in R} t_i^{rh} Y_i^{rh} \forall j \in E\backslash\{O\}, \forall i \in E_{(j)}^- \tag{11}$$

$$X_{ij}^r = \{0,1\} \,\forall (i,j) \in A, \forall r \in R_{ij} \tag{12}$$

$$Y_i^{rh} = \{0,1\} \,\forall i \in E, \forall r,h \in R_i \tag{13}$$

The objective function (1) minimizes the total cost, including transportation cost and transit cost. Objective function (2) represents to minimize total transportation time, including transportation and transit time. Constraint (3) ensures flow conservation; Constraint (4) states that at most one transportation mode is selected between two nodes; Constraint (5) represents that each transit node should perform mode transformation at most once. Constraints (6) ~ (7) guarantee the compatibility among decision variables, achieving the rationality and integrity of node-arc continuity. Constraint (8) represents the total carbon emission is equal to the sum of the carbon emissions resulting from transportation and transit; Constraint (9) represents capacity constraint between two nodes; Constraint (10) guarantees the transit capacity of each node. Constraint (11) represents that relationship between the arrival times of two adjacent nodes along the transportation route. Constraints (12) ~ (13) define 0–1 decision variables.

The features of MOLCMTPP-FDFT can be summarized as illustrated in Fig. 1. After inputting the parameters ( such as unit distance cost, distance, carbon emission allowance, unit carbon cost, carbon tax, and unit carbon emission ) included in MOLCMTPP-FDFT, the non-dominated solutions/multimodal transportation routes that satisfy constraints (3)-(13) and minimize the objective functions (1) and (2) are obtained using reasonable solution method, which are the outputs of MOLCMTPP-FDFT.

The MOLCMTPP-FDFT considered in this paper is a combinatorial problem. Similar to other well-known combinatorial optimization problems, such as basic multimodal transportation planning problem (Chen, & Wang, 2016), vehicle routing problem (Zhang, Zhang, Ma, & Liu, 2019), and assignment problem, the feasible solution set of the proposed mathematical model (1)-(13) is the intersection of constraints (3)-(13). That is to say, its boundary conditions are implicated in its constraints. In addition, if the demand is deterministic and fuzzy time intervals are not considered, it becomes deterministic model. This can also be considered as a boundary condition.

### 3.2. The MOLCMTPP-FDFT under different carbon policies

The low-carbon multimodal transportation model with fuzzy demand and fuzzy time is transformed under mandatory carbon emissions, carbon tax, carbon trading and carbon offset. These are respectively referred to as Model II, Model III, Model IV and Model V. The following subsections provide a detailed description.

#### 3.2.1. The model under mandatory carbon emission policy – model II

The government's mandated regulations on carbon emission require that logistics transportation should be conducted strictly in compliance with the carbon emission allowance. Therefore, Model II retains the objectives and constraints in Model I, while ensuring that the maximum carbon emissions cannot be exceeded under this policy.

$$Q_{\text{emission}} \le Q_M \tag{14}$$

Constraint (14) means that the total carbon emission must be less than the upper bound of emission stipulated by the government.

#### 3.2.2. The model under carbon tax policy – model III

Carbon tax is defined as the cost associated with carbon emissions, which includes a fixed tax rate $\gamma$ multiplied by total carbon emissions and added to the total cost $Z_1$. Therefore, in Model III, the objective function should incorporate the cost of carbon emissions into the total

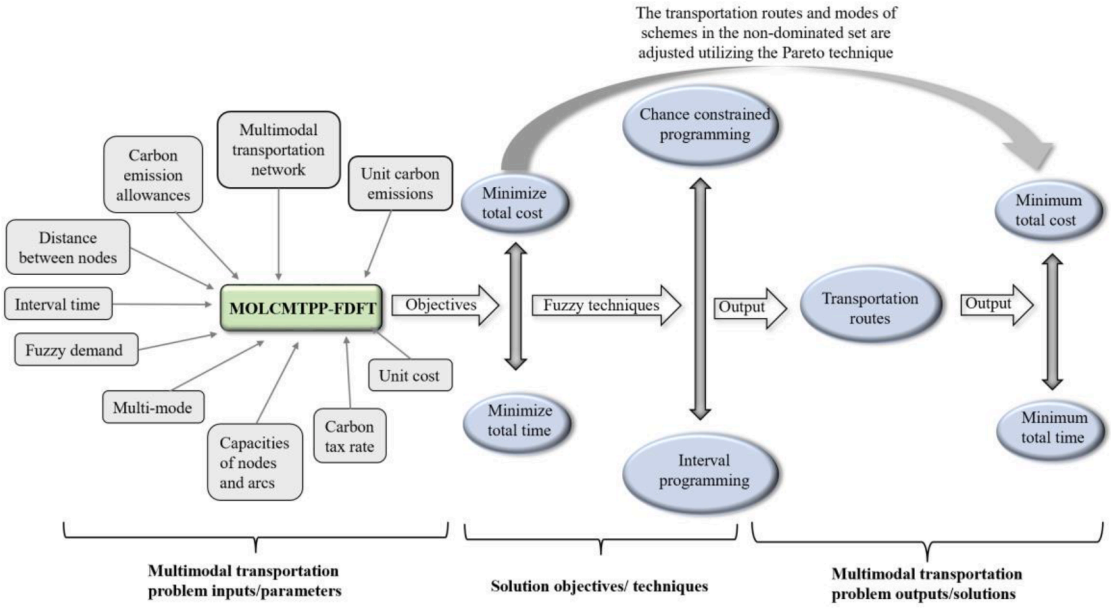



**Fig. 1.** Summary of the inputs, outputs and objectives of the MOLCMTPP-FDFT.

cost using equation (15), while still satisfying constraints (4) ~ (13).

$$\min Z_{\text{tax}} = Z_1 + \gamma Q_{\text{emission}} \tag{15}$$

### 3.2.3. The model under carbon trading policy – model IV

Carbon trading refers to the practice of enterprises holding a specific amount of carbon emission allowances, which can be bought or sold externally based on actual transportation needs in order to minimize costs while ensuring successful transportation. The cost of carbon trading should be taken account into the total cost $Z_1$, while objective 2 $Z_2$ and other constraints in Model I are preserved. Model IV is formulated as follows:

$$\min Z_{\text{trading}} = Z_1 + c_{\text{trading}} Q_{\text{trading}} \tag{16}$$

s.t

$$Q_{\text{trading}} + Q_{\text{emission}} = Q_T \tag{17}$$

$$Q_T \geq 0 \tag{18}$$

Objective (16) represents the minimum total cost considering the carbon trading cost, where $c_{\text{trading}} Q_{\text{trading}}$ is the product of unit price of carbon trading and the volume of carbon trading. Constraint (17) indicates that the carbon emission allowance is composed of actual emissions and volume of carbon trading. Constraint (18) means that the carbon emissions of the purchased carbon trading are non-negative. Meanwhile, constraints (4) ~ (13) are satisfied.

### 3.2.4. The model under carbon offset policy – model V

Carbon offset defines that enterprises can purchase allowance from external sources when the existing carbon emission allowance is insufficient for transportation purposes. However, if the enterprise has the remaining allowances of carbon emission after its transportation, which can be sold to other companies. In other words, enterprises must bear the cost for the excess emissions if exceeding the regulated carbon emission allowance; whereas if its emissions are within the regulatory allowance, which cannot be traded. The total cost $Z_1$ of objective (1) need to consider the cost of carbon offset, which means objective (1) will be transformed into equation (19) while objective 2 and other constraints of Model I are not changed. Model V is constructed as follows:

$$\min Z_{\text{offset}} = Z_1 + c_{\text{offset}} Q_{\text{offset}} \tag{19}$$

s.t

$$Q_{\text{offset}} = \begin{cases} Q_{\text{emission}} - Q_O, Q_{\text{emission}} \geq Q_O \\ 0, Q_{\text{emission}} < Q_O \end{cases} \tag{20}$$

Objective (19) represents minimizing the total cost, including the costs of transportation, transit and carbon offset. The carbon offset cost is the product of unit cost of carbon offset and the amount of carbon offset. Constraint (20) indicates that the amount of carbon offset is determined by the relationship between the actual carbon emissions and the amount of quota $Q_O$ under carbon offset. Meanwhile, constraints (4) ~ (13) are satisfied.

## 4. The fuzzy approach

Currently, stochastic programming, fuzzy programming and interval programming are commonly used to deal with uncertain parameters in routing optimization. Interval programming belongs to the category of fuzzy programming. In generally, less data can be used to evaluate the probability distribution of demand for the decision makers, resulting in a less feasible stochastic programming. Existing literature commonly employs triangular fuzzy variables, trapezoidal fuzzy variables, and interval numbers as methods to represent those uncertain variables. The maximum membership degree of triangular fuzzy numbers can only correspond to one value, which is only applicable to a clearly defined range. In contrast, trapezoidal fuzzy numbers are more flexible, and the maximum membership degree of trapezoidal fuzzy numbers can correspond to an interval, which allows different decision makers to hold varying opinions on the most likely value. Some scholars have proposed that the trapezoidal fuzzy numbers can better represent the fuzzy demand in actual situation. In addition, weather, road conditions and personnel habits all impact transportation time. It is possible to roughly determine the time range for each road section and transit node based on past experience and recent road conditions. Therefore, the flexible trapezoidal fuzzy number is used to represent fuzzy demand while using interval numbers for dealing fuzzy time.

### 4.1. General fuzzy measure

The General Fuzzy (GF) measure, proposed by Xu and Zhou (2011), is utilized to quantify the probability of fuzzy phenomena. Let $(\Theta, P(\Theta),$

*Pos* ) denote the possibility space, where $\mathscr{B}$ represents an event in $P(\Theta)$, and $Pos\{\mathscr{B}\}$ indicates the possibility of event $\mathscr{B}$ occurring. Moreover, the necessity of event $\mathscr{B}$ occurring can be determined as $Nec\{B\} = 1 - Pos\{\mathscr{B}^C\}$, where $\mathscr{B}^C$ represents the complement of event $\mathscr{B}$. The general fuzzy measure can be expressed as a convex combination of possibility (*Pos*) and necessity (*Nec*), yielding $GF\{\mathscr{B}\} = Nec\{\mathscr{B}\} + \mathscr{L}(Pos\{\mathscr{B}\} - Nec\{\mathscr{B}\}) = \mathscr{L}Pos\{\mathscr{B}\} - (1-\mathscr{L})Nec\{\mathscr{B}\}$. The parameter $\mathscr{L}(0 \le \mathscr{L} \le 1)$ serves as a determinant of the decision maker's inclination towards optimism or pessimism (Nasim, Mohsen, Hosseinzadeh, & Hasan, 2022; Xu & Zhou, 2011, 2013).

Moreover, triangular and trapezoidal fuzzy variables for handling uncertain demand is more suitable since they are better aligned with the actual problem description (Dong, & Wan, 2018). Additionally, Zhang (2022) provides several reasons why trapezoidal fuzzy variables outperform triangular ones. Firstly, the trapezoidal fuzzy variables are allowed to have multiple most probable values, which meet the actual situation that different decision makers and experts have different opinions on the most probable values. Secondly, the uncertain parameters are expressed as trapezoidal fuzzy variables to make the decision more flexible. Finally, by significantly reducing the upper and lower bounds of the fuzzy variable, trapezoidal fuzzy variables can be conveniently approximated as triangular fuzzy variables (Zhang, 2022). In practical scenarios, the uncertain demand is often represented as a trapezoidal fuzzy variable $\widetilde{q} = (q_1, q_2, q_3, q_4)$, where $q_1, q_4$ denote the upper and lower bounds of demand respectively, and $[q_2, q_3]$ is the most likely interval of demand. These four values are generally obtained from previous data or experiences. The membership function $\mu_{\widetilde{q}}(q)$ is illustrated in Fig. 2 and the possibility, necessity and general fuzzy measure are expressed as Eq. (21) ~ (23), respectively, where $\lambda$ represents the crisp demand.

$$Pos(\widetilde{q} \le \lambda) = \begin{cases} 1 & \lambda \ge q_2 \\ \dfrac{\lambda - q_1}{q_2 - q_1} & q_1 \le \lambda \le q_2 \\ 0 & \text{otherwise} \end{cases} \tag{21}$$

$$Nec(\widetilde{q} \le \lambda) = \begin{cases} 1 & \lambda \ge q_4 \\ \dfrac{\lambda - q_3}{q_4 - q_3} & q_3 \le \lambda \le q_4 \\ 0 & otherwise \end{cases} \tag{22}$$

$$GF(\widetilde{q} \le \lambda) = \begin{cases} 1 & \lambda \ge q_4 \\ \mathscr{L} + (1-\mathscr{L})\left(\dfrac{\lambda - q_3}{q_4 - q_3}\right) & q_3 \le \lambda \le q_4 \\ \mathscr{L} & q_2 \le \lambda \le q_3 \\ \mathscr{L}\left(\dfrac{\lambda - q_1}{q_2 - q_1}\right) & q_1 \le \lambda \le q_2 \\ 0 & \lambda \le q_1 \end{cases} \tag{23}$$

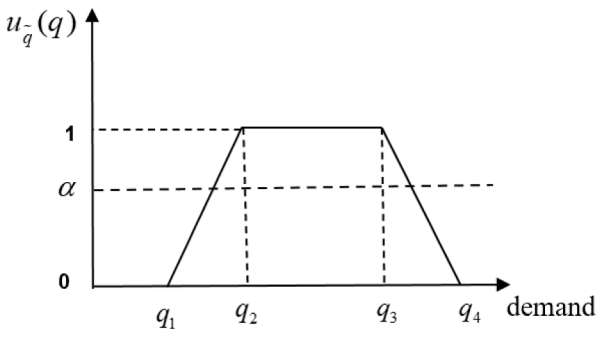



**Fig. 2.** Membership function of trapezoidal fuzzy variable.

According to the properties of fuzzy general measures, the fuzzy chance constraints can be expressed in a crisp form when $\mathscr{L}$ is less than or greater than a specific confidence level $\alpha$, which is shown as Eq. (24):

$$GF\{\widetilde{q} \le \lambda\} \ge \alpha \Leftrightarrow \begin{cases} \left(\dfrac{\mathscr{L} - \alpha}{\mathscr{L}}\right) q_1 + \left(\dfrac{\alpha}{\mathscr{L}}\right) q_2 \le \lambda, \alpha \le L \\ \left(\dfrac{1-\alpha}{1-\mathscr{L}}\right) q_3 + \left(\dfrac{\alpha - \mathscr{L}}{1-\mathscr{L}}\right) q_4 \le \lambda, \alpha \ge L \end{cases} \tag{24}$$

In $GF\{\mathscr{B}\} = \mathscr{L}Pos\{\mathscr{B}\} - (1-\mathscr{L})Nec\{\mathscr{B}\}$, if $\mathscr{L}$ is equal to 1, 0, and 0.5, then the *GF* measure represents possibility, necessity, and credibility, respectively (Nasim, Mohsen, Hosseinzadeh, & Hasan, 2022; Peykani, Mohammadi, Emrouznejad, Pishvaee, & Rostamy-Malkhalifeh, 2019; Peykani, Hosseinzadeh, Sadjadi, Ebrahimnejad, & Mohammadi, 2021). The variable $\mathscr{L}$ in the *GF* measure represents the decision-maker's inclination towards meeting expectations (Zhou, & Xu, 2020; Peykani, Mohammadi, & Emrouznejad, 2020; Qiao, & Liu, 2019). In order to maintain the objectivity of decision-makers regarding events, the paper defines $\mathscr{L}$ as 0.5. Consequently, as depicted in Eq. (25), the *GF* measure is equivalent to credibility ($C_r$).

$$GF(\widetilde{q} \le \lambda) = C_r(\widetilde{q} \le \lambda) = \begin{cases} 1 & \lambda \ge q_4 \\ \dfrac{q_4 - 2q_3 + \lambda}{2(q_4 - q_3)} & q_3 \le \lambda \le q_4 \\ \dfrac{1}{2} & q_2 \le \lambda \le q_3 \\ \dfrac{\lambda - q_1}{2(q_2 - q_1)} & q_1 \le \lambda \le q_2 \\ 0 & \lambda \le q_1 \end{cases} \tag{25}$$

According to Eq. (25), $GF(\lambda \ge \widetilde{q}) \ge \alpha$ implies that $\lambda \ge 2\alpha q_2 + (1-2\alpha)q_1$ when $\alpha \in [0, 0.5]$. $GF(\lambda \ge \widetilde{q}) \ge \alpha$ denotes that $\lambda \ge (2\alpha - 1)q_4 + 2(1-\alpha)q_3$ if $\alpha \in [0.5, 1]$. The uncertain variable $\widetilde{q}$ involved in the objective function and constraints can be converted into the deterministic variable in Eq. (26) at different confidence levels.

$$\widetilde{q} = \begin{cases} 2\alpha q_2 + (1-2\alpha)q_1 & 0 \le \alpha \le 0.5 \\ (2\alpha - 1)q_4 + 2(1-\alpha)q_3 & 0.5 \le \alpha \le 1 \end{cases} \tag{26}$$

### 4.2. *Interval programming*

Interval programming (Dong, & Wan, 2018; Charnes, & Collomb, 1972) is a fuzzy mathematical programming model used to solve uncertain variables represented by interval numbers, which is widely applied in real-world situations. Its related theories and methods have been widely studied (Sun, Gong, Zeng, & Geng, 2018). Interval programming model refers to objective functions or constraints involve uncertain variables represented as interval number. Let $a^L$ and $a^U$ represent the upper and lower bounds of the interval number, respectively, and the minimum interval programming model can be mathematically formulated as (27).

$$\begin{aligned} &\min f = \sum_{i=1}^{n} [a^L, a^U] x_i \\ &\text{s.t} \\ &\begin{cases} \sum_{i=1}^{n} [a^L, a^U] x_i \ge [b^L, b^U] \\ [a^L, a^U] x_i = [b^L, b^U] \\ x_i \ge 0, i = 1, 2, 3, \cdots, n \end{cases} \end{aligned} \tag{27}$$

The transportation time in each arc and the unit transit time at random node are represented as interval numbers in this paper, denoted as $[t_{ijr}^L, t_{ijr}^U]$ and $[t_{irh}^L, t_{irh}^U]$, respectively. Combining with the interval number optimization model, the maximum and minimum values of the transportation time objective function are represented as (28) and (29), respectively.

$$Z_2^{\min} = \sum_{(i,j)\in A}\sum_{r\in R} t_{ijr}^L X_{ij}^r + \sum_{i\in E\setminus\{O,D\}}\sum_{r\in R}\sum_{h\in R} t_{irh}^L \widetilde{q} Y_i^{rh} \tag{28}$$

$$Z_2^{\max} = \sum_{(i,j)\in A}\sum_{r\in R} t_{ijr}^U X_{ij}^r + \sum_{i\in E\setminus\{O,D\}}\sum_{r\in R}\sum_{h\in R} t_{irh}^U \widetilde{q} Y_i^{rh} \tag{29}$$

The objective function $Z_2$ is defined as $Z_2 = [Z_2^{\min}, Z_2^{\max}]$, and for any value $Z_2^{'} \in [Z_2^{\min}, Z_2^{\max}]$, the risk coefficient $\varepsilon$ can be calculated as follows:

$$\varepsilon = \frac{Z_2^{'} - Z_2^{\min}}{Z_2^{\max} - Z_2^{\min}}, \varepsilon \in [0,1] \tag{30}$$

$$Z_2^{'} = \varepsilon Z_2^{\max} + (1-\varepsilon) Z_2^{\min} \tag{31}$$

The decision maker can obtain the shortest total transportation time value $Z_2^{'}$ under the premise of assuming the risk coefficient $\varepsilon$. Therefore, the uncertain objective (2) in the original MOLCMTPP-FDFT model can be transformed into a deterministic objective function, as shown in (32):

$$\begin{aligned} \min Z_2 = {} & \varepsilon\left(\sum_{(i,j)\in A}\sum_{r\in R} t_{ijr}^U X_{ij}^r + \sum_{i\in E\setminus\{O,D\}}\sum_{r\in R}\sum_{h\in R} t_{irh}^U \widetilde{q} Y_i^{rh}\right) \\ & + (1-\varepsilon)\left(\sum_{(i,j)\in A}\sum_{r\in R} t_{ijr}^L X_{ij}^r + \sum_{i\in E\setminus\{O,D\}}\sum_{r\in R}\sum_{h\in R} t_{irh}^L \widetilde{q} Y_i^{rh}\right) \end{aligned} \tag{32}$$

Where $\varepsilon$ denotes the degree of risk that the decision maker is willing to take in order to obtain the optimal objective value.

### 4.3. *Chance constrained programming*

Chance constrained programming (Charnes, & Cooper, 1959) is an effective method proposed by Charnes in 1959 to deal with uncertain parameters in practical environment. Fuzzy chance constrained programming is a mathematical programming model based on fuzzy sets and fuzzy measure theory to solve uncertain variables involved in constraints or objective functions. Its principle is to ensure that the probability of satisfying constraints is not lower than the confidence level $\alpha \in [0,1]$ of fuzzy variables (Huang, 2005). Fuzzy chance constrained programming can be converted into deterministic demand through confidence level. The multi-objective chance constrained programming model is illustrated as (33).

$$\begin{aligned} & \min[\overline{f}_1, \overline{f}_2, \cdots, \overline{f}_k] \\ & \text{s.t} \\ & \begin{cases} GF\{f_i(x,\xi) \le \overline{f}_i\} \ge \delta_i, i = 1,2,\cdots,k \\ GF\{g_j(x,\xi) \le 0\} \ge \beta_j, j = 1,2,\cdots,p \\ \overline{f}_i = \min\{GF\{f_i(x,\xi) \le \overline{f}_i\} \ge \delta_i\} \end{cases} \end{aligned} \tag{33}$$

Where $x$ is the decision variable, $\xi$ defines the fuzzy variable, and $f_i(x,\xi)$ represents the $i$th objective function. $g_j(x,\xi)$ denotes the $j$th constraint; $\delta_i$, $\beta_j$ are the predefined confidence levels, respectively; $GF\{\}$ denotes the fuzzy measure of $\{\}$.

In this paper, the demand $\widetilde{q} = (q_1, q_2, q_3, q_4)$ means that uncertainty parameter represented by trapezoidal fuzzy variables. The confidence levels for the objective function are assumed to be $\alpha$, while $\beta$ represents the confidence level of the constraints. Consequently, the objectives and constraints involving fuzzy demand in MOLCMTPP-FDFT under different low-carbon policies can be transformed into fuzzy chance constrained programming models, which is expressed as (34).

$$\begin{aligned} & \min\overline{Z}_1, \min\overline{Z}_2 \\ & \text{s.t} \\ & \begin{cases} GF\{Z_1 \le \overline{Z}_1\} \ge \alpha \\ GF\{Z_2 \le \overline{Z}_2\} \ge \alpha \\ GF\left\{\sum_{(i,j)\in A}\sum_{r\in R} \widetilde{q} e_r d_{ij}^r X_{ij}^r + \sum_{i\in E\setminus\{O,D\}}\sum_{r\in R}\sum_{h\in R} \widetilde{q} e_{rh} Y_i^{rh} \le Q\right\} \ge \beta \\ GF\left\{\Omega_{ij}^r\left(X_{ij}^r - 1\right) \le \widetilde{q} \le \Omega_{ij}^r X_{ij}^r\right\} \ge \beta \\ GF\left\{\Psi_i^{rh}\left(Y_i^{rh} - 1\right) \le \widetilde{q} \le \Psi_i^{rh} Y_i^{rh}\right\} \ge \beta \end{cases} \end{aligned} \tag{34}$$

## 5. The ATDSSA for MOLCMTPP-FDFT

### 5.1. *Sparrow search algorithm (SSA)*

The solution space in SSA is explored through the simulation of sparrows' foraging behavior. The superior individual within the population is referred to as the producers, while the remaining individuals are known as scroungers. Producers are responsible for procuring food for the entire population and guiding the foraging direction. In comparison to scroungers, producers possess a broader search range. If any sparrows detect danger, they will promptly update their corresponding location in order to mount a counterattack. Xue and Shen (2020) demonstrate that sparrows possess the ability to flexibly switch between their roles as producers and scroungers. Eq. (35) and Eq. (36) illustrate the updates of location for these two identities.

$$X_{i,j}^{t+1} = \begin{cases} X_{i,j}^t \bullet \exp\left(\dfrac{-i}{\omega \bullet iter_{\max}}\right), R_1 \le ST \\ X_{i,j}^t + U \bullet L, R_1 \ge ST \end{cases} \tag{35}$$

$$X_{i,j}^{t+1} = \begin{cases} U \bullet \exp\left(\dfrac{X_{\text{worst}}^t - X_{i,j}^t}{i^2}\right), i \le M/2 \\ X_P^{t+1} + \left|X_{i,j}^t - X_P^{t+1}\right| A^+ \bullet L, \text{othersise} \end{cases} \tag{36}$$

Where $X_{i,j}^t$ and $X_{\text{worst}}^t$ denote the location of the $i$th sparrow and the worst location in the $t$th iteration, respectively. $X_{i,j}^{t+1}$ and $X_P^{t+1}$ are the location of the $i$th sparrow and the location of the best individual in the $(t+1)$th iteration, respectively. $iter_{\max}$ represents the maximum number of iterations; $\omega$ is a random number in (0,1]; $R_1 \in [0,1]$ and $ST \in [0.5,1]$ represent the warning value and safety threshold, respectively. $U$ is a random number with a normal distribution. $L$ is a $d \times 1$ unit column vector. $A$ represents $1 \times d$ row vector ($d$ is the encoding length), where the elements are given 1 or $-1$ at random, and $A^+ = A^T(AA^T)^{-1}$, $A^T$ is the transpose matrix of $A$; $M$ is the population size. The detailed iterative optimization process is shown in reference (Xue, & Shen, 2020).

### 5.2. *t Distribution*

$t$ distribution, also known as the student distribution (Guerrero-Cusumano, 1996), is characterized by a parameter called the degrees of freedom $n$. The shape of its curve is dependent on the value of $n$. The probability density function is calculated using Eq. (37), with the curve becoming flatter and the middle of the curve lower as $n$ decreases, while the bilateral tail of the curve becomes higher.

$$p(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{n\pi} \times \Gamma\left(\frac{n}{2}\right)}\left(1 + \frac{x^2}{2}\right)^{-\frac{n+1}{2}} \tag{37}$$

where gamma function $\Gamma\left(\frac{n+1}{2}\right) = \int_0^{+\infty} x^{\frac{n+1}{2}-1} e^{-x} dx$. The $t$ distribution approaches to a Gaussian distribution as $n \to \infty$, and it converges to the Cauchy distribution as $n \to 1$. The Gaussian and Cauchy distributions are two specific instances of the $t$ distribution.

### 5.3. Sparrow search algorithm with adaptive t distribution (ATDSSA)

ATDSSA incorporates adaptive $t$ distribution into SSA, and its innovation point is that the number of iterations is used as the degree of freedom parameter of $t$ distribution to perturb the sparrow location. In the initial iterations, the mutation of the $t$ distribution is equivalent to that of the Cauchy distribution, which has better global search ability. As the iterations progress, the degrees of freedom of $t$ distribution variation increase, which is similar with Gaussian distribution and has good local exploration ability. The number of iterations was utilized to adjust the degrees of freedom in the $t$ distribution, achieving a balance between local and global search and enhancing the optimization performance of the algorithm. Moreover, the non-dominated sorting and mutation operations in NSGA-II are also employed to solve the proposed mathematical model of MOLCMTPP-FDFT and improve the optimization performance of ATDSSA. The individual locations are updated adaptively by $t$ distribution using Eq. (38).

$$X_i^* = X_i + X_i \times t(T) \tag{38}$$

The $t$ distribution is utilized to enhance the SSA. Firstly, the algorithm parameters are configured, and the initial population is randomly generated. The individual objective function value of the population is calculated, and the Pareto solution set is selected. The population individuals are categorized as producers and scroungers based on the pareto front. Additionally, the updates for the locations of producers and scroungers are performed separately. The $t$ distribution mutation operation is performed on the population when the density value of the $t$ distribution exceeds a random number. The loop terminates when the maximum number of iterations is reached, and the optimal solution set is output. The specific process of the adaptive $t$ distribution sparrow search algorithm is illustrated in Fig. 3.

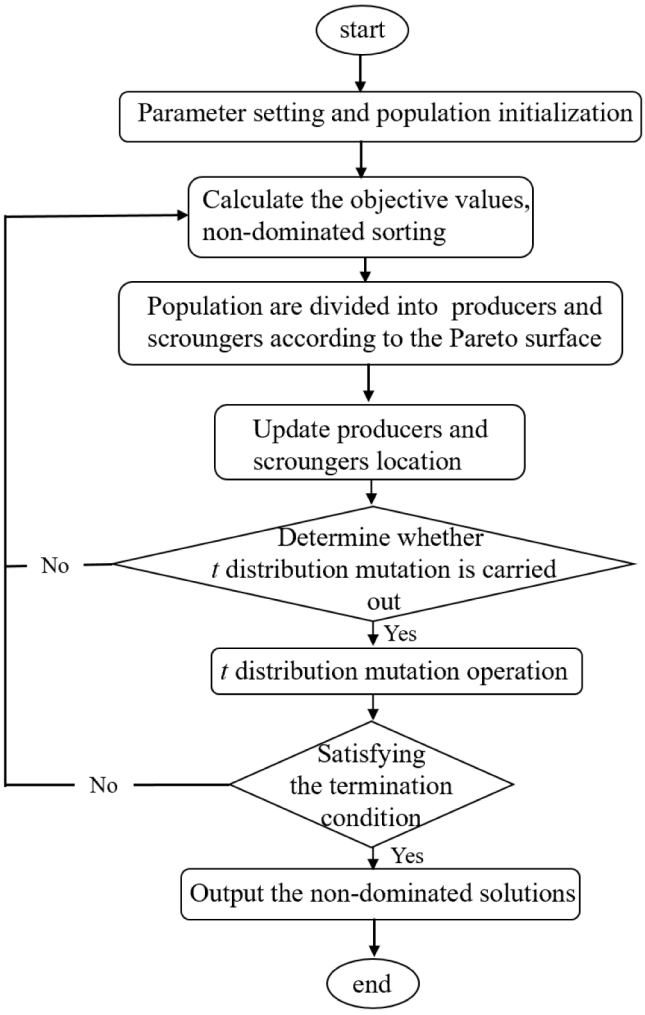



**Fig. 3.** The process of ATDSSA algorithm.

#### 5.3.1. Solution representation

Multimodal transportation planning problem involves the selection of transit nodes, determining whether mode conversion should take place at these nodes, and choosing the appropriate transportation mode between two given nodes. To address this problem, natural number coding is used in this article with each individual's length being $2n-1$. The first part is the route code with a length of $n$, where the largest and smallest natural numbers are the original node and destination of the multimodal transportation route, respectively. The traversed nodes are represented by their corresponding natural numbers, and the untraversed nodes are replaced by 0. The second part is the encoding of the transportation mode, with a length of $n-1$, where 1, 2 and 3 denote road, rail and sea, respectively. For the purpose of clarity, Fig. 4. illustrates the encoding for a feasible solution in a multimodal transportation network with $n = 10$.

Part 1 in Fig. 4 illustrates the route code that is represented by zero and natural numbers, 1, 2, ……., and 10, and its length is equal to the number of nodes in the multimodal transportation network. The route code presented in part 1 shows that the transportation route starts from the original node 1 and then passes through nodes 3, 6 and 8 to the destination node 10. Since nodes 2, 4, 5, 7, and 9 are not included in the transportation route, the elements at these positions are equal to 0. Part 2 illustrates the transportation modes departing from each node, excluding the destination node 10 that does not have a corresponding transportation mode. The numbers 1, 2, and 3 denote three distinct transportation modes. According to the route code presented in part 1 and the elements at positions 1,3, 6 and 8 in part 2, it is easy to know that the multimodal transportation route should be original node 1 $\xrightarrow{\text{mode 1}}$ node 3 $\xrightarrow{\text{mode 3}}$ node 6 $\xrightarrow{\text{mode 2}}$ node 8 $\xrightarrow{\text{mode 1}}$ destination node 10. The transportation route does not pass through nodes 2, 4, 5, 6, 7, and 9. However, for the sake of operational convenience in subsequent processes, a suitable transportation mode is still randomly assigned to each of these nodes. For example, transportation mode 1 is assigned to node 2, so the second element in part 2 is 1.

#### 5.3.2. Fast construction method of Pareto non-dominated solution

Non-dominated sorting method is a common solution strategy for multi-objective optimization problems, and it has been described in detail in many literatures. This paper adopted the fast non-dominated sorting algorithm (FNSA) proposed by Deb in 2002 to solve the multi-objective problem (Deb, Agrawal, Pratap, & Meyarivan, 2002). Deb, Agrawal, Pratap and Meyarivan (2002) provided a comprehensive definition and detailed explanation of FNSA. The FNSA is extensively employed in the renowned Non-dominated Sorting Genetic Algorithm II (NSGA-II). FNSA partitions the merged population into multiple layers of non-dominated fronts, which furnish crucial individual-level information for generating population of the succeeding generation and accomplishing the selection operation. The time complexity of FNSA is $O(NM^2)$, where $N$ represents the number of objective functions, and $M$ denotes the population size. Algorithm 1 presents the pseudocode for FNSA. Firstly, the non-dominated solution set $I'$ and dominated solution set $I''$ are initially defined as empty sets (line 2). Lines 3–18 depict the process of non-dominated sorting for all individuals within the population. The individual $I_i$ added to the dominant solution set $I''$ if there

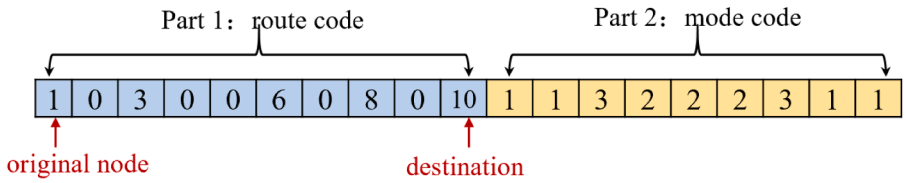



**Fig. 4.** Representation of an individual.

exists an individual $I_j$ that dominates $I_i$ ($I_j \prec I_i \& I_j \neq I_i$) (lines 4–14). Otherwise, individual $I_i$ is added to the non-dominated solution set $I'$ (line 16), and this process is repeated until all individuals are assigned to either the non-dominated or dominated solution set. The non-dominated solution set and the dominated solution set are merged into a new set $I^*$ by prioritizing the non-dominated solutions. Then, those sets $I^*$, $I'$ and $I''$ are subsequently generated as output (lines 19–20).

**Algorithm 1**:Pseudocode of the Fast Non-dominated Sorting

1: **function** $[I^*, I', I'] =$ fast nondominated sorting $(I^M, M)$ // $I^M$– Set of population individuals, $I_i, I_j \in I^M$, $M$– population size
2: $I' = \varnothing; I'' = \varnothing$
3: **for** $i := 1$ to $M$ **do**
4: $flag = 0$
5: **for** $j := 1$ to $M$ **do**
6: **if** $I_j \neq I_i$ **then**
7: **if** $I_j \prec I_i$ **then** // $I_j$ dominated $I_i$
8: $flag = 1$
9: **break**
10: **end if**
11: **end if**
12: **end for**
13: **if** $flag = 1$ **then**
14: $I' := [I', I_i]$
15: **else**
16: $I' = [I', I_i]$
17: **end if**
18: **end for**
19: $I^* = [I', I'']$
28: **return** $I^*, I', I''$
29: **end function**

### 5.3.3. The location update process

The original SSA is well-suited for solving continuous optimization problems. Given that the problem studied in this paper is discrete, we propose an improvement to the location update mechanism for producers and scroungers, with specific steps outlined as follows:

*Step* 1: An individual $I_{\text{gbest}}$ randomly selected from the non-dominated set and a producer (scrounger) $I_j$ are chosen as parents to update the location of $I_j$ by implementing single-point crossover operation on their transportation routes/modes if $R_1 \leq ST$($ST$ represents the warning value). Otherwise, a feasible solution is randomly generated.

*Step* 2: randomly select a transit node/ mode in the parents $I_j$ and $I_{\text{gbest}}$ as the crossover points, respectively, and implement route/mode crossover to generate offspring individuals $I_{\text{child1}}$ and $I_{\text{child2}}$.

*Step* 3: adjust the preceding and subsequent nodes of the changed node, as well as their transportation route/mode, based on the multimodal transportation network to ensure the feasibility of $I_{\text{child1}}$ and $I_{\text{child2}}$.

*Step* 4: calculate the objectives of $I_{\text{child1}}$ and $I_{\text{child2}}$, and perform non-dominated comparison between each offspring and $I_j$, respectively. If non-dominated solution is found in the offspring, it will be preserved. Otherwise, $I_j$ retains constant.

**Algorithm 2**:Pseudocode of the update process

1: **function** $I^* =$ update_process $(I_j, I_{\text{gbest}}, ST, A)$ // *A represents multimodal transportation network*
2: $N =$ find$(I_j(1:\text{length}(I_j)) = 0)$;
3: $R_1 =$ rand
4: **if** $R_1 \leq ST$ **do**

(*continued on next column*)

(*continued*)

**Algorithm 2**:Pseudocode of the update process

5: $a =$ sort$(N($randperm$(\text{length}(N), 2))$
6: **while** $a(1) = 1$ **do**
7: $a =$ sort$(N($randperm$(\text{length}(N), 2))$
8: **end while**
9: $b1 = I_j(1:a(1)-1); b2 = I_{\text{gbest}}(a), I_{\text{gbest}}(a); b3 = I_j(a(2)+1, \text{end})$;
10: $I_{\text{child}} = [b1 b2 b3]$
11: **else**
12: $I_{\text{child}} =$ initialization$(A)$
13: **end if**
14: $I_{\text{child}} =$ feasible $(I_{\text{child}})$
15: **if** $I_{\text{child}} \prec I_j$ **do**
16: $I^* = I_{\text{child}}$
17: **elseif** $I_j \prec I_{\text{child}}$ **then**
18: $I^* = I_j$
19: **else**
20: $I^*(1,:) = I_j; I^*(2,:) = I_{\text{child}}$;
21: **return** $I^*$
22: **end function**

Algorithm 2 shows the pseudocode for updating individual $I_j$, where $I_{\text{gbest}}$ represents an individual in the first non-dominated front; $ST$ denotes the safety threshold, and $A$ states the inter-node relationships within the multimodal transportation network. The specific procedure is as follows: Firstly, the individual $I_j$ is identified to traverse those numbered nodes, and the warning value $R_1$ is randomly generated (lines 2 ~ 3). Two nodes different from the origin node are then selected at random as the starting and end nodes for route/mode across a section of the network if $R_1 \leq ST$. Otherwise, a new individual is randomly generated as an offspring (lines 4 ~ 13). The feasibility of offspring individual $I_{\text{child}}$ is monitored (line 14). Subsequently, $I^*$ is updated (lines 15 ~ 21).

### 5.3.4. t Distribution mutation

The degree of freedom of the $t$ distribution is determined by the number of iterations, and its density function value $P(t)$ is compared with a random number rand. If $P(t) >$ rand, a node different from the destination is randomly selected from the updated route and $t$ distribution variation is performed, Otherwise, a feasible solution is generated randomly. The steps for $t$ distribution mutation are as follows:

*Step* 1: calculate the local objectives of different transportation modes from the mutation node to all forward nodes, respectively.

*Step* 2: the two objective values are weighted linearly to form the total fitness cost, fitness $= Z_1 + Z_2 \times c_{\text{time}}$, where $c_{\text{time}}$ means unit time cost.

*Step* 3: select the forward node and transportation mode that have the minimum local fitness as the forward local route of the mutation node.

*Step* 4: take the selected forward node as the new mutation node, and repeat *Steps* 1 ~ 3 until the mutation node is the destination to end the cycle.

In order to facilitate understanding, Fig. 5. presents a solution in which one node conducts a $t$ distribution mutation process. In Fig. 5., $C^r_{(a,b)}$ represents the fitness value of transportation mode $r$ on arc $(a,b)$, and $C_a$ represents the fitness from origin node to the mutation node $a$. As shown in Fig. 5., the mutation node, node 3, has 3 forward nodes including nodes 5, 6, and 7. There are 3, 2 and 3 transportation modes

from node 3 to it forward nodes 5, 6 and 7, respectively. Among them, transportation mode 1 from nodes 3 to 7 has the minimal local fitness.

## 6. Case study

In order to verify the effectiveness of the improved algorithm for solving the proposed model, this paper conducted computational experiments on a real multimodal transportation network in China. This real case involves 15 cities and 3 modes. Matlab 2020a was used for the implementation of this algorithm. Those experiments were carried out on a personal laptop equipped with AMD Ryzen 7 5800U with Radeon Graphics, CPU,1.90 GHz, 16.0 GB RAM under Windows 11 professional system. The primary data source for the experiment is YTO Express Co., LTD, and the reference (Zhang, 2022) is utilized. The algorithm parameters were initialized based on the initial values mentioned in reference (Li & Wang, 2022), and they were subsequently updated through parameter tuning and Taguchi method.

### 6.1. Problem statement and data issues

The multimodal network comprises 15 cities including Nanchang, Guiyang, Chongqing, Nanchang, Changsha, Wuhan, Hefei, Shanghai, Xuzhou, Jinan, Zhengzhou, Taiyuan, Beijing, Dalian and Harbin, which are donated as O, 1, 2… 13 and D. Table 2 shows the distance between each pair of cities by sea, rail, and road modes. Table 3 lists the fuzzy arrival time interval of each arc, and the transit time interval at each node.

Table 4 presents the values of other parameters such as unit cost $c_r$, unit mode transit cost $c_{rh}$, unit transport carbon emission $e_r$ and unit mode transit carbon emission $e_{rh}$, which are set according to the literature Zhang, Jin, Yuan, & Zhang, 2021. In addition, the mandatory carbon emissions allowance is set as 20t $(1t = 1 \times 10^3 \mathrm{kg})$ based on the allocation principle of carbon emission allowance in the national carbon trading market. Carbon trading and offset policies are also established at a limit of 10t $(1t = 1 \times 10^3 \mathrm{kg})$, with a unit cost of 7.8 for both policies. The unit rate for carbon tax is set at 10.

### 6.2. Parameter setting and sensitivity analysis in ATDSSA

The quality of the solutions generated by ATDSSA can be tremendously influenced by several parameters, including the population size $M$, the number of iterations $T$, risk coefficient $\varepsilon$ and the confidence levels of $\alpha$and$\beta$. These parameters play a crucial role in controlling the search strategies employed. To ensure repeatability and consistency in our results, we have kept these parameters constant throughout the optimization process. This approach is commonly referred to as parameter tuning in literature Zhang, Zhang, Ma, & Liu, 2019 and Zhang, Liu, Ma, & Zhang, 2020. In order to maintain reproducibility, all experiments were conducted using a predefined set of parameter settings obtained through parameter tuning techniques (Zhang, Zhang, Ma, & Liu, 2019; Zhang, Liu, Ma, & Zhang, 2020).

To verify the influence of these parameters on the performance of ATDSSA, we conducted an analysis of approximate optimal solutions and key parameters on a multimodal transportation network consisting

**Table 2**
Distance between two nodes under different transportation modes (unit: km).

| $(i,j)$ | Distance road | rail | sea | $(i,j)$ | Distance road | rail | sea |
|---|---|---|---|---|---|---|---|
| ( O, 1 ) | 604 | 875 | 105 | ( 6, 9 ) | 675 | 614 | – |
| ( O, 2 ) | 986 | 1338 | 422 | ( 7, 10 ) | 942 | 998 | – |
| ( 1, 3 ) | 1156 | 1264 | 337 | ( 7, 11 ) | 1356 | 1497 | – |
| ( 1, 4 ) | 793 | 949 | – | ( 7, 12 ) | 1206 | 1328 | – |
| ( 2, 5 ) | 932 | 1159 | 658 | ( 8, 10 ) | 366 | 349 | – |
| ( 2, 6 ) | 1285 | 1492 | 1322 | ( 8, 11 ) | 775 | 926 | – |
| ( 3, 7 ) | 728 | 807 | – | ( 8, 12 ) | 695 | 814 | – |
| ( 3, 8 ) | 743 | 817 | – | ( 9, 10 ) | 446 | 668 | – |
| ( 3, 9 ) | 1056 | 1162 | – | ( 9, 11 ) | 526 | 529 | 497 |
| ( 4, 7 ) | 1066 | 1173 | 398 | ( 9, 12 ) | 410 | 495 | 1026 |
| ( 4, 8 ) | 993 | 1247 | – | ( 10, D ) | 1976 | 2139 | – |
| ( 4, 9 ) | 1181 | 1291 | – | ( 11, 12 ) | 491 | 568 | 612 |
| ( 5, 7 ) | 821 | 811 | – | ( 11, 13 ) | 1304 | 1452 | 1168 |
| ( 5, 8 ) | 641 | 595 | 572 | ( 11, D ) | 1774 | 1846 | – |
| ( 5, 9 ) | 846 | 976 | – | ( 12, 13 ) | 840 | 938 | 697 |
| ( 6, 7 ) | 437 | 457 | – | ( 12, D ) | 1288 | 1278 | – |
| ( 6, 8 ) | 318 | 295 | – | ( 13, D ) | 1032 | 946 | – |

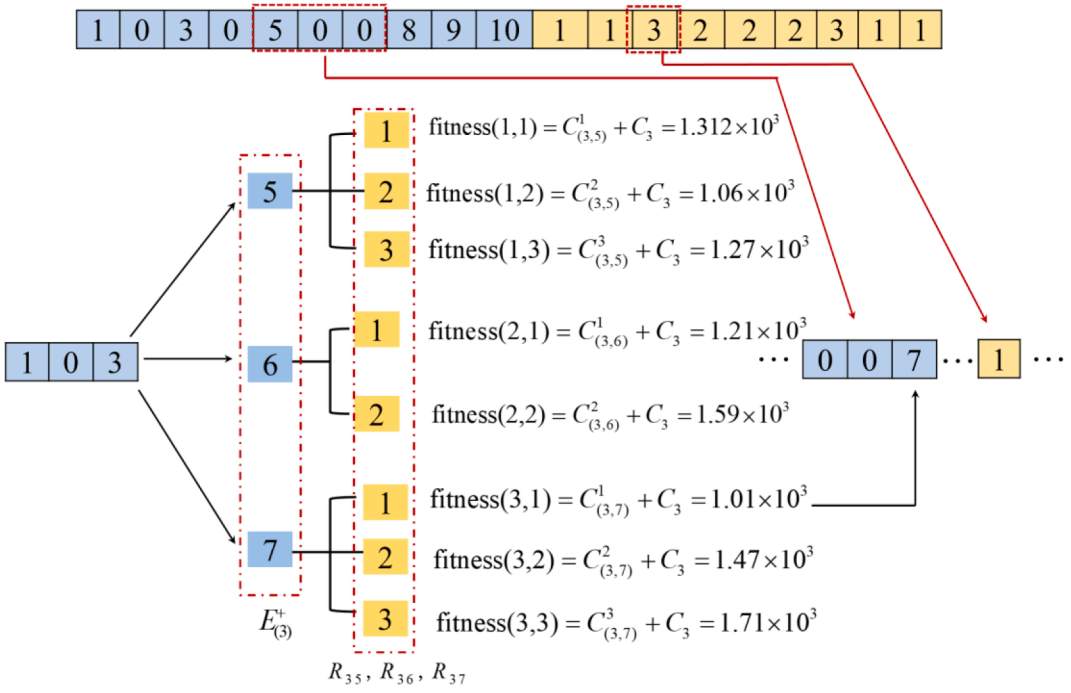



**Fig. 5.** $t$ distribution mutation.

**Table 3**
Interval of time between two nodes under different transportation modes (unit: day).

| $(i,j)$ | Time interval | | | $(i,j)$ | Time interval | | |
|---|---|---|---|---|---|---|---|
| | road | rail | sea | | road | rail | sea |
| ( O, 1 ) | [2.7,3] | [3.2,3.5] | [0.4,0.8] | ( 6, 9 ) | [3.1,3.4] | [2.3,2.5] | – |
| ( O, 2 ) | [4.5,4.9] | [5,5.4] | [1.4,1.5] | ( 7, 10 ) | [4.3,4.7] | [3.7,4] | – |
| ( 1, 3 ) | [5.3,5.8] | [4.7,5.1] | [1.1,1.2] | ( 7, 11 ) | [6.2,6.8] | [5.5,6] | – |
| ( 1, 4 ) | [3.6,4] | [3.5,3.8] | – | ( 7, 12 ) | [5.5,6] | [4.9,5.3] | – |
| ( 2, 5 ) | [4.2,4.7] | [4.3,4.6] | [2.2,2.4] | ( 8, 10 ) | [1.7,1.8] | [1.3,1.4] | – |
| ( 2, 6 ) | [5.8,6.4] | [5.5,6] | [4.4,4.7] | ( 8, 11 ) | [3.5,3.9] | [3.4,3.7] | – |
| ( 3, 7 ) | [3.3,3.6] | [3,3.2] | – | ( 8, 12 ) | [3.2,3.5] | [3,3.3] | – |
| ( 3, 8 ) | [3.4,3.7] | [3,3.3] | – | ( 9, 10 ) | [2,2.2] | [2.5,2.7] | – |
| ( 3, 9 ) | [4.8,5.3] | [4.3,4.6] | – | ( 9, 11 ) | [2.4,2.6] | [2,2.1] | [1.7,1.8] |
| ( 4, 7 ) | [4.8,5.3] | [4.3,4.7] | [1.3,1.4] | ( 9, 12 ) | [1.9,2.1] | [1.8,2] | [3.4,3.7] |
| ( 4, 8 ) | [4.5,5] | [4.6,5] | – | ( 10, D ) | [9,9.9] | [7.9,8.6] | – |
| ( 4, 9 ) | [5.4,5.9] | [4.8,5.2] | – | ( 11, 12 ) | [2.2,2.5] | [2.1,2.3] | [2,2.2] |
| ( 5, 7 ) | [3.7,4.1] | [3,3.2] | – | ( 11, 13 ) | [5.9,6.5] | [5.4,5.8] | [3.9,4.4] |
| ( 5, 8 ) | [2.9,3.2] | [2.2,2.4] | [1.9,2] | ( 11, D ) | [8.1,8.9] | [6.8,7.4] | – |
| ( 5, 9 ) | [3.8,4.2] | [3.6,3.9] | – | ( 12, 13 ) | [3.8,4.2] | [3.5,3.8] | [2.3,2.5] |
| ( 6, 7 ) | [2.2,2.4] | [1.7,1.8] | – | ( 12, D ) | [5.9,6.4] | [4.7,5.1] | – |
| ( 6, 8 ) | [1.4,1.6] | [1.1,1.2] | – | ( 13, D ) | [4.7,5.2] | [3.5,3.8] | – |

**Table 4**
Parameters of different transportation modes.

| Mode | $c_r$(¥/(km.kg)) | $c_{rh}$(¥/kg) | | | $e_r$(kg/(km.kg)) | $e_{rh}$(kg/kg) | | |
|---|---|---|---|---|---|---|---|---|
| | | road | rail | sea | | road | rail | sea |
| road | 0.262 | 0 | 8 | 9 | 0.0440 | 0 | 0.128 | 0.117 |
| rail | 0.491 | 8 | 0 | 10 | 0.0127 | 0.128 | 0 | 0.113 |
| sea | 0.462 | 9 | 10 | 0 | 0.0091 | 0.117 | 0.113 | 0 |

of 35 nodes. In studying the relationship between a parameter (or a pair) and its effect on approximate optimal solutions, this instance was independently tested for 10 runs while keeping other parameters constant (Zhang, Zhang, Ma, & Liu, 2019; Zhang, Liu, Ma, & Zhang, 2020).

#### 6.2.1. Population size

The optimization performance of ATDSSA is significantly impacted by the population size $M$. In this study, the following evaluation index (Hisao, Tadashi, & Tadahiko, 2003) is employed to reasonably set the parameter $M$.

$$V_{NDS}(S_i) = \frac{|S_i - \{p_1 \in S_i | \exists p_2 \in S : p_2 \prec p_1\}|}{|S_i|} \tag{39}$$

where $S_M$ represents the non-dominated solutions obtained for the given population size $M$. $S$ is the set of all non-dominated solutions obtained by different population sizes. $p_2 \prec p_1$ means that $p_2$ dominates $p_1$. $|S_M|$ donates individual numbers in $S_M$. The average value of $V_{NDS}$ was defined as $AVG$. A larger $AVG$ indicates a better effect of parameter $M$

We calculated $AVG$ values for 10 different population sizes. The largest AVG value is observed in Table 5 when $M = 6n$, where $n$ is the number of nodes in the multimodal transportation network. Therefore, the population size $M$ is set as $6n$ in this study.

#### 6.2.2. Number of iterations

The number of iterations in ATDSSA plays a crucial role. Excessive iterations lead to inefficient utilization of computational resources, while too few iterations terminate the search process prematurely and fail to yield an optimal/ near-optimal solution (Zhang, Liu, Ma, & Zhang, 2020). Table 6 shows that results for instances over 10 runs. It includes the average number of iterations to obtain approximate non-dominated solution set, the number of non-dominated solutions $N_{non}$, average objectives $Z_{1-AVG}$ and $Z_{2-AVG}$ across all non-dominated solutions, average carbon emission $EM$ and average CPU time in seconds. As shown in Table 6, the non-dominated solutions are basically stable when T $\geq$ 200, and 200 is set as the maximum number of iterations to save computation time.

#### 6.2.3. Taguchi method for designing parametersM, T

The purpose of the Taguchi method is to identify the parameter combination that ensures algorithmic stability and minimizes volatility (Taguchi, 1986). The orthogonal table is utilized for the selection and arrangement of experimental parameters, enabling the evaluation of various factors affecting experimental results through fewer parameter combinations (Taguchi, 1986). Both controllable factors parameters and noise factors (random errors) have an impact on the experimental results. The impact of noise factors on experimental results can be reduced through adjusting the level of controllable parameters. The stability of the experimental results was evaluated using the Signal-to-Noise Ratio ($SNR$), which was computed using equations (36) ~ (37), with higher values indicating higher quality. Based on the maximum $SNR$, the best value for each parameter is determined. For a more comprehensive understanding of the Taguchi method, interested readers may refer to Taguchi (1986), Taguchi, Chowdhury and Wu (2005).

As shown in Table 7, 4 different values are initially set for parameters $M$ and $T$, respectively. Thus, 16 different combinations in the orthogonal table (Table 8) were generated based on the values of parameters $M$ and $T$. For each parameter combination, its $SNR$ were obtained using Eq. (40) and the experimental results $f_i (i = 1, 2, 3)$ of 3 runs. In Eq. (36), $k$ is the number of runs. $f_i$ is calculated using Eq. (41), where $\overline{Z}_1$ and $\overline{Z}_2$ are respectively the averages of objective 1 and objective 2 of all the non-dominated solutions obtained in the $i$th run, and $Z_1^{\min}$ and $Z_2^{\min}$ are respectively the minimum value of objective 1 and objective 2 among all non-dominated solutions. In Eq. (41), $f$ represents the weighted average of normalized objectives, which is generally used to evaluate the fitness of non-dominated solutions. The advantage of this approach is that it can avoid the bias of different orders of magnitude and is easy to computationally implement.

$$SNR = -10\log\frac{1}{k}\sum_{i=1}^{k}\frac{1}{f_i^2} \tag{40}$$

$$f = \frac{\overline{Z}_1}{Z_1^{\min}} + \frac{\overline{Z}_2}{Z_2^{\min}} \tag{41}$$

**Table 5**
Corresponding *AVG* for different values of..*n*

| *M* | *n* | 2*n* | 3*n* | 4*n* | 5*n* | 6*n* | 7*n* | 8*n* | 9*n* | 10*n* |
|---|---|---|---|---|---|---|---|---|---|---|
| *AVG* | 0.017 | 0.102 | 0.341 | 0.341 | 0.240 | 0.642 | 0.341 | 0.462 | 0.240 | 0.462 |

**Table 6**
Average results of different scale instances over 10 runs obtained using different number of iterations.

| *T* | $N_{non}$ | $X_1$ | | $X_2$ | | $Z_{1-AVG}$ | $Z_{2-AVG}$ | *EM* | CPU time(s) |
|---|---|---|---|---|---|---|---|---|---|
| | | $Z_1$ | $Z_2$ | $Z_1$ | $Z_2$ | | | | |
| 100 | 3 | 358957.92 | 23.40 | 385906.80 | 19.62 | 375505.84 | 21.60 | 17847.13 | 59.87 |
| 120 | 4 | 356917.68 | 21.64 | 440870.16 | 16.10 | 388491.96 | 19.50 | 14347.98 | 76.35 |
| 140 | 3 | 325298.40 | 19.78 | 440870.16 | 16.10 | 364228.00 | 17.87 | 13223.25 | 93.99 |
| 160 | 3 | 274295.28 | 17.92 | 326515.44 | 17.72 | 299280.00 | 17.82 | 15712.11 | 126.84 |
| 180 | 4 | 274295.28 | 17.92 | 440870.16 | 16.1 | 338901.36 | 17.15 | 12550.13 | 170.01 |
| 200 | 5 | 260730.48 | 17.93 | 394205.52 | 13.04 | 313934.26 | 15.46 | 13738.91 | 201.34 |
| 220 | 5 | 260730.48 | 17.93 | 394205.52 | 13.04 | 313934.26 | 15.46 | 13738.91 | 253.91 |
| 240 | 5 | 260730.48 | 17.93 | 394205.52 | 13.04 | 313934.26 | 15.46 | 13738.91 | 291.76 |
| 260 | 5 | 260730.48 | 17.93 | 394205.52 | 13.04 | 313934.26 | 15.46 | 13738.91 | 340.34 |
| 280 | 5 | 260730.48 | 17.93 | 394205.52 | 13.04 | 313934.26 | 15.46 | 13738.91 | 390.76 |

**Table 7**
The levels of different parameters.

| Parameters | Level 1 | Level 2 | Level 3 | Level 4 |
|---|---|---|---|---|
| Population size *M* | 4*n* | 5*n* | 6*n* | 7*n* |
| Number of iterations *T* | 180 | 200 | 220 | 240 |

Table 9 shows the value of *SNR* for each parameter at different level. Taking parameter *M* as an example, *M* is level 2 in combinations 5,6,7 and 8. Therefore, the *SNR* of parameter *M* at level 2 is equal to the average of *SNR* obtained in combinations 5,6,7 and 8. From Table 9, it is easy to know that the maximum *SNR* s are obtained when $M = 6n$ and $T = 200$. respectively.

### 6.2.4. Sensitivity analysis of risk coefficientε

The risk coefficient ε in the context of fuzzy interval programming is a stochastic ranking measure ranging from 0 to 1, which reflects the decision maker's tolerance for risk when implementing transportation schemes under uncertain conditions. The model is executed for each risk level of parameter ε to determine the minimum possible output value, and a real case is evaluated and analyzed using the proposed ATDSSA at each risk level. Table 10 presents ten different risk levels for ε between (0,1] with a step size of 0.1, as this parameter primarily effects on objective 2. The table displays average results over 10 runs, including the minimum value $Z_{1-\min}$, the average value $Z_{1-AVG}$, and the maximum value $Z_{2-\max}$ for objective 2 among non-dominated solution set.

As shown in Fig. 6, parameter ε is positively correlated with the number of time intervals, and larger ε results in larger objective 2 when the actual time is closer to the fuzzy time upper bound.

### 6.2.5. Sensitivity analysis of confidence levels α andβ

The confidence level $\alpha$ is determined based on the horizontal cut method of the membership function as described in reference Fazayeli, Eydi, & Kamalabadi, 2018. The value of $\alpha$ varies from 0.1 to 1 with a step size of 0.1. The average results of 10 runs of the test instances are presented in Table 11. It includes the average values of objective $Z_{1-AVG}$ and $Z_{2-AVG}$, as well as the corresponding solutions $X_1$ and $X_2$ for the maximum and minimum objectives among non-dominated solutions. Additionally, it provides the average carbon emission *EM* and CPU time in seconds. Equation (22) shows that $q_3$ and $q_4$ have a greater impact on higher confidence level $\alpha$. This means that the higher the value of the demand pair membership function, the higher the demand and the higher the cost (Fazayeli, Eydi, & Kamalabadi, 2018). Similarly, the value of the demand function is lower and the cost decreases for lower

**Table 8**
The orthogonal table under the Taguchi method.

| Combination | Parameter level | | Experimental results | | | *SNR* |
|---|---|---|---|---|---|---|
| | *M* | *T* | $f_1$ | $f_2$ | $f_3$ | |
| 1 | 1 | 1 | 2.1341 | 2.0000 | 2.0745 | 6.3082 |
| 2 | 1 | 2 | 2.1227 | 2.1104 | 2.0146 | 6.3648 |
| 3 | 1 | 3 | 2.1039 | 2.1337 | 2.1206 | 6.5238 |
| 4 | 1 | 4 | 2.0781 | 2.1221 | 2.1104 | 6.4580 |
| 5 | 2 | 1 | 2.1470 | 2.1412 | 2.0781 | 6.5325 |
| 6 | 2 | 2 | 2.1271 | 2.1470 | 2.1470 | 6.6095 |
| 7 | 2 | 3 | 2.1341 | 2.0781 | 2.1312 | 6.5021 |
| 8 | 2 | 4 | 2.1312 | 2.0781 | 2.1377 | 6.5068 |
| 9 | 3 | 1 | 2.1470 | 2.1470 | 2.1212 | 6.6014 |
| 10 | 3 | 2 | 2.1470 | 2.1470 | 2.1470 | 6.6366 |
| 11 | 3 | 3 | 2.1470 | 2.1312 | 2.1377 | 6.6026 |
| 12 | 3 | 4 | 2.1312 | 2.1470 | 2.0781 | 6.5191 |
| 13 | 4 | 1 | 2.1341 | 2.0781 | 2.1341 | 6.5059 |
| 14 | 4 | 2 | 2.1377 | 2.1470 | 2.1470 | 6.6240 |
| 15 | 4 | 3 | 2.1470 | 2.1270 | 2.0210 | 6.4282 |
| 16 | 4 | 4 | 2.0000 | 2.1377 | 2.1470 | 6.4094 |

**Table 9**
The average *SNR* across different parameters and levels.

| Parameters | Level 1 | Level 2 | Level 3 | Level 4 |
|---|---|---|---|---|
| Population size *M* | 6.4137 | 6.5377 | **6.5899** | 6.4919 |
| Number of iterations *T* | 6.4870 | **6.5588** | 6.5142 | 6.4733 |

levels of $\alpha$ (Fazayeli, Eydi, & Kamalabadi, 2018). As shown in Fig. 7, subfigure (a) shows the effect on objective 1, which increases slowly with increasing of $\alpha \in (0, 0.5]$ and increases largely with increasing of $\alpha \in (0, 0.5]$. In subfigure (b), it is obvious that objective 2 increases slowly if $\alpha \leq 0.6$. However, objective 2 changes obviously when $\alpha \in [0.6, 0.8]$. Therefore, it can be inferred that different schemes can be obtained for various confidence levels $\alpha$ based on the decision maker's selection of an appropriate preference value. The experimental results illustrate that the confidence level $\alpha$ has an equal impact on cost and carbon emissions. The following analyses will focus on the most sensitive phase of $\alpha$, which is defined as 0.7.

Additionally, the influence of $\beta$ on fuzzy demand is mainly on constraints, and has little effect on the two objectives. $\beta$ in constraints (9) ~ (10) affects the selection of transit nodes. In this paper, the capacity of transit nodes and road segments is inviolable. Therefore, the confidence level $\beta$ of the constraints is defined as 1 to ensure that the capacity is not violated.

**Table 10**
Average results of different scale instances over 10 runs obtained using different number of iterations.

| ε | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|
| $Z_{2-\min}$ | 12.86 | 13.04 | 13.23 | 13.41 | 13.59 | 13.77 | 13.95 | 14.14 | 14.32 | 14.50 |
| $Z_{2-AVG}$ | 15.2 | 15.46 | 15.72 | 15.98 | 16.24 | 16.5 | 16.76 | 17.02 | 17.28 | 17.54 |
| $Z_{2-\max}$ | 17.59 | 17.93 | 18.28 | 18.62 | 18.97 | 19.32 | 19.66 | 20.01 | 20.35 | 20.7 |

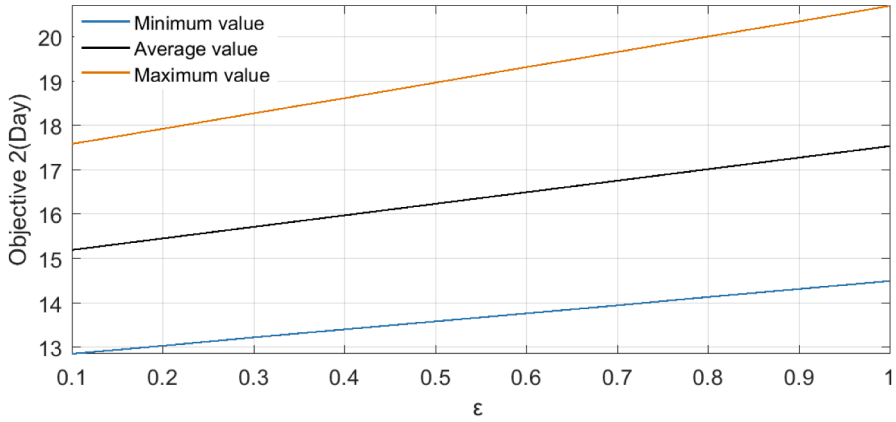



**Fig. 6.** Objective 2 with respect to credibility parameter $\varepsilon$.

**Table 11**
Average results of different scale instances over 10 runs obtained using different number of iterations.

| $\alpha$ | $N_{non}$ | $X_1$ | | $X_2$ | | $Z_{1-AVG}$ | $Z_{2-AVG}$ | $EM$ | CPU time(s) |
|---|---|---|---|---|---|---|---|---|---|
| | | $Z_1$ | $Z_2$ | $Z_1$ | $Z_2$ | | | | |
| 0.1 | 4 | 102015.76 | 14.69 | 180677.53 | 12.71 | 144057.07 | 13.84 | 8592.65 | 198.66 |
| 0.2 | 4 | 111289.92 | 14.73 | 197102.76 | 12.73 | 157153.17 | 13.91 | 9373.80 | 186.01 |
| 0.3 | 4 | 111289.92 | 14.73 | 197102.76 | 12.73 | 157153.17 | 13.91 | 9373.80 | 213.33 |
| 0.4 | 4 | 129838.24 | 14.81 | 229953.22 | 12.78 | 183345.37 | 14.04 | 10936.10 | 205.54 |
| 0.5 | 4 | 139112.40 | 14.86 | 246378.45 | 12.81 | 196441.46 | 14.10 | 11717.25 | 208.11 |
| 0.6 | 3 | 204031.52 | 15.14 | 361355.06 | 12.99 | 294285.27 | 14.36 | 12196.82 | 194.32 |
| 0.7 | 5 | 260730.48 | 17.93 | 394205.52 | 13.04 | 313934.26 | 15.46 | 13738.91 | 201.34 |
| 0.8 | 4 | 297153.22 | 15.97 | 427055.98 | 13.10 | 354504.80 | 14.99 | 13062.06 | 187.09 |
| 0.9 | 4 | 320011.16 | 16.14 | 459906.44 | 13.15 | 381774.40 | 15.14 | 13787.73 | 203.51 |

### 6.3. Effectiveness of the proposed t distribution mutation in ATDSSA

The effectiveness of the $t$ distribution mutation update strategy in ATDSSA is verified in this subsection. To determine whether incorporating a $t$ distribution mutation operation in solving the MOLCMTPP-FDFT problem enhances the performance of ATDSSA, the optimization algorithm SSA1 is employed, which is used to solve the aforementioned instance while keeping other parameters and factors unchanged.

SSA1: the number of iterations serves as the degree of freedom for the $t$ distribution, which is utilized to update the population's location during the counterattack phase when an individual is at risk.

See Fig. 8. shows the non-dominated solutions obtained using ATDSSA and SSA1 after 10 runs, respectively. When the non-dominated frontier is closer to the lower left corner, it indicates a higher quality of solutions. As shown in Fig. 8, it is evident that non-dominated solutions obtained by ATDSSA are closer to the lower left corner than those obtained using SSA1, which indicates that utilizing the number of iterations as a degree of freedom to update population locations during danger encounters is an effective strategy that enhances effectiveness of ATDSSA.

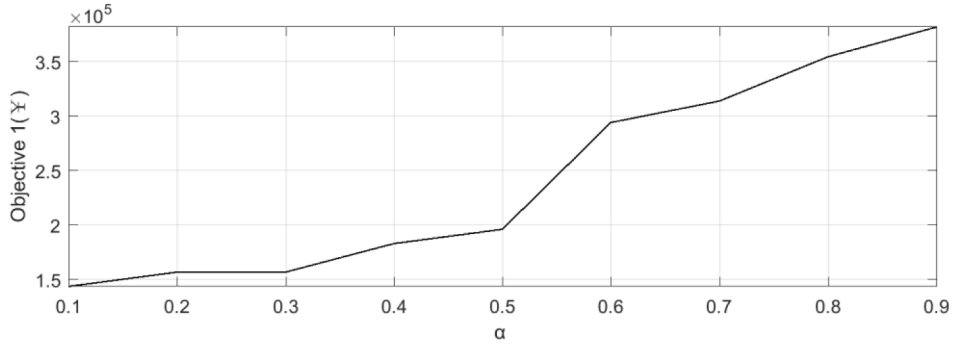
### 6.4. Comparison and analysis with other intelligent algorithms

In order to verify the performance of the proposed ATDSSA, we conducted a comparative analysis with other state-of-the-art algorithms such as grey wolf optimization (GWO) (Mirjalili, Mirjalili, & Lewis, 2014), gravity search algorithm, (GSA) (Rashedi, Nezamabadi-pour, & Saryazdi, 2009), particle swarm optimization (PSO) (Zhang, Liu, Ma, & Zhang, 2020) and NSGA-II (Deb, Agrawal, Pratap, & Meyarivan, 2002). The results obtained using each algorithm under carbon tax policy are shown in Table 12, which includes representative non-dominated solutions $I_{gbest1}$ with the minimum cost and $I_{gbest2}$ with minimum transportation time, average cost $Z_{1-AVG}$, average transportation time $Z_{2-AVG}$, average carbon emission $Q_{AVG}$ of the non-dominant solutions, and CPU time in seconds.

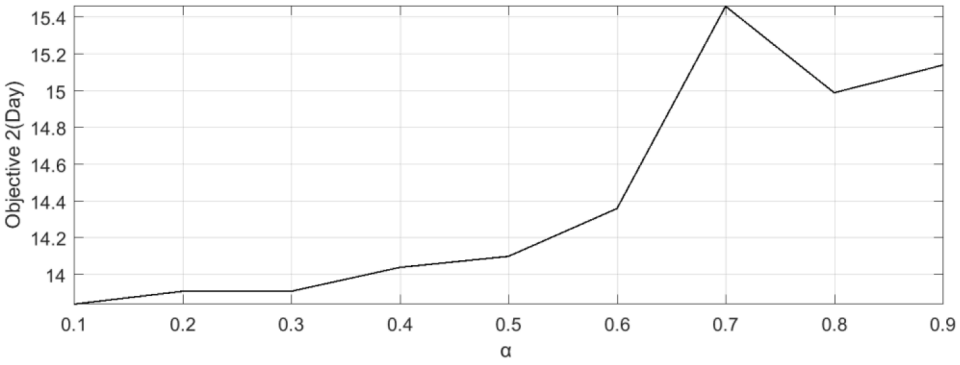
From Table 12, it is easy to know that the non-dominated solutions obtained by GWO, GSA, PSO and SSA are inferior to those obtained by ATDSSA. Although both ATDSSA and NSGA-II can obtain better solutions, ATDSSA needs less computation time while ensuring the same high-quality solutions. The results presented in Table 12 not only indicate that the ATDSSA is a very competitive algorithm for solving the multimodal transportation planning problem, but also demonstrate that the $t$ distribution mutation operation effectively improves the performance of SSA.

### 6.5. Experimental results under different low-carbon policies

The proposed ATDSSA is also used to solve the MOLCMTPP-FDFT under different low-carbon policies including mandatory carbon emission, carbon tax, carbon trading, and carbon offset. The non-dominated




(a) Average values of objective 1 with respect to parameter $\alpha$




(b) Average values of objective 2 with respect to parameter $\alpha$

**Fig. 7.** Average values of objective 1 and 2 with respect to credibility parameter.$\alpha$

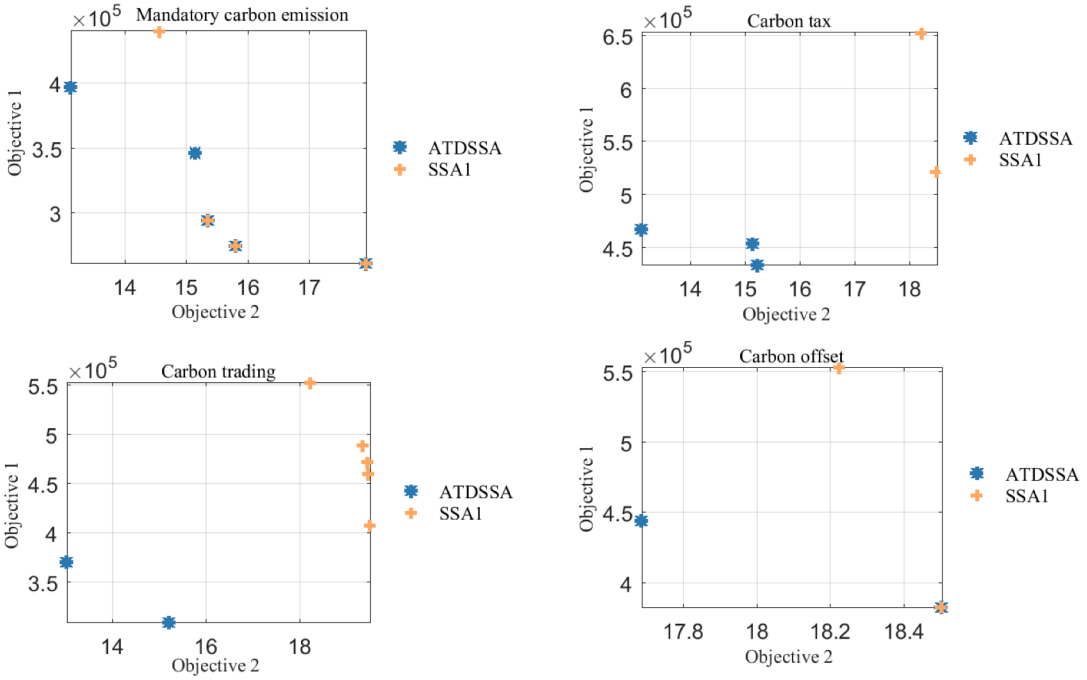



**Fig. 8.** The non-dominant solutions of $t$ distribution mutation under different carbon policies.

solutions and their corresponding average values obtained under each low-carbon policy are presented in Table 13. Comparing the solutions shown in Table 13, it is easy to observe that the solutions obtained under carbon tax policy have the lowest carbon emission, but their transportation costs are also significantly higher, which indicates that while the implementation of a carbon tax policy can effectively reduce carbon emissions, it may also lead to a largely increase in transportation costs. Carbon offset has the weakest impact on the reduction of carbon emission, and the solutions obtained under it have higher transportation costs. The emission reduction effect of mandatory carbon emission and

**Table 12**
The non-dominated solutions obtained by different algorithms under carbon tax policy.

| Algorithm | $I_{gbest1}$ | | $I_{gbest2}$ | | $Z_{1-AVG}$ | $Z_{2-AVG}$ | $Q_{AVG}$ | CPU time(s) |
|---|---|---|---|---|---|---|---|---|
| | $Z_1$ | $Z_2$ | $Z_1$ | $Z_2$ | | | | |
| GWO | 433521.28 | 15.22 | 47774.72 | 14.67 | 455207.67 | 15.01 | 13002.94 | **73.95** |
| GSA | 454327.00 | 15.15 | 47774.72 | 14.67 | 466050.86 | 14.91 | 15006.23 | 172.10 |
| PSO | 514336.44 | 17.87 | 570021.64 | 13.97 | 545160.58 | 15.62 | 17301.22 | 214.77 |
| NSGA-II | **433521.28** | **15.22** | **467665.85** | **13.12** | **451838.04** | **14.50** | **12977.83** | 220.01 |
| SSA | 521186.61 | 18.50 | 652914.52 | 18.22 | 587050.57 | 18.36 | 18847.50 | 87.65 |
| ATDSSA | **433521.28** | **15.22** | **467665.85** | **13.12** | **451838.04** | **14.50** | **12977.83** | 102.31 |

carbon trading lies between that of carbon tax and carbon offset. Although the transportation costs and carbon emissions of solutions obtained using mandatory carbon emission are slightly lower than those obtained using carbon trading, their transportation time is comparatively longer.

The current practice in the logistics industry is that many companies still rely solely on a single- mode transportation, either by road or rail. For example, the well-known logistics companies in China, YTO and ZTO. To further confirm that the mathematical model of MOLCMTPP-FDFT proposed in this study is indeed helpful for enterprises to reduce emissions, the transportation cost $Z_1$, transportation time $Z_2$ and carbon emission *EM* associated with the single-mode transportation routes, both road and rail, are also calculated. Table 14 presents the results under different low-carbon policies.

**Table 13**
Experimental results.

| Type of carbon emission policy | Number of schemes | Route | $Z_1$ | $Z_2$ | *EM* |
|---|---|---|---|---|---|
| Mandatory carbon emission | 1 | O-S-1-S-3-H-9-R-12-H-D | 260730.48 | 17.932 | 19349.01 |
| | 2 | O-S-1-S-3-R-8-H-12-H-D | 274295.28 | 15.808 | 17321.49 |
| | 3 | O-S-1-S-3-H-8-H-12-R-D | 294109.92 | 15.348 | 14240.87 |
| | 4 | O-S-1-S-3-H-8-R-12-R-D | 346330.08 | 15.148 | 10799.69 |
| | 5 | O-S-1-S-3-R-9-R-12-R-D | 397269.36 | 13.124 | 7039.649 |
| | **Average values** | | **314547.02** | **15.472** | **13750.14** |
| Carbon tax | 1 | O-S-1-S-3-H-8-H-12-H-D | 433521.28 | 15.224 | 21094.14 |
| | 2 | O-S-1-S-3-H-8-R-12-R-D | 454327.00 | 15.148 | 10799.69 |
| | 3 | O-S-1-S-3-R-9-R-12-R-D | 467665.85 | 13.124 | 7039.649 |
| | **Average values** | | **451838.04** | **14.499** | **12977.83** |
| Carbon trading | 1 | O-S-1-S-3-H-8-H-12-H-D | 309114.16 | 15.224 | 21094.14 |
| | 2 | O-S-1-S-3-R-8-R-12-R-D | 370676.94 | 13.044 | 6983.515 |
| | **Average values** | | **339895.55** | **14.134** | **14038.83** |
| Carbon offset | 1 | O-S-1-H-4-H-9-H-12-H-D | 382359.06 | 18.504 | 27648.89 |
| | 2 | O-S-1-S-3-H-8-R-11-R-12-R-D | 444150.21 | 17.688 | 12267.81 |
| | **Average values** | | **413254.64** | **18.096** | **19958.35** |

As shown in Table 13, the solutions obtained by MOLCMTPP-FDFT under different low-carbon policies are different. The average costs, time, and carbon emissions for non-dominated solutions obtained from MOLCMTPP-FDFT are superior to those of single-mode transportation under different low-carbon policies in Table 14, respectively. In Table 13, the transportation cost of solutions obtained under the carbon offset policy are 413254.64, which is much higher than that of solutions obtained under the carbon trading policy. The main reason is that enterprises can sell excess carbon emission allowances to reduce their costs under the carbon trading policy. Under mandatory carbon emission and carbon tax, solutions have lower average carbon emissions that are 13750.14 and 12977.83, respectively, which are conductive to achieve the purpose of protecting the environment. In the initial stage of low-carbon implementation, carbon trading can partially alleviate transportation costs, thereby facilitating the promotion and adoption of low-carbon transportation practices. The solutions obtained under carbon offsets have slightly higher cost, which indicates that the carbon offsets should be implemented at the expense of transportation cost. The carbon offset policy can be integrated with other low-carbon policies to effectively address the limitations of individual policies. The proposed MOLCMTPP-FDFT is advantageous in terms of reducing logistics costs and alleviating environmental burdens, thereby facilitating the achievement of low-carbon transportation.

**Table 14**
The actual single-mode transportation.

| Type of carbon emission policy | Mode | Route | $Z_1$ | $Z_2$ | *EM* |
|---|---|---|---|---|---|
| Mandatory carbon emission | Road | O–H-1-H-4-H-9-H-12-H-D | 268874.88 | 19.88 | 31984.48 |
| | Rail | O-R-1-R-4-R-9-R-12-R-D | 576001.92 | 18.32 | 10553.19 |
| | **Average values** | | **422438.40** | **19.100** | **21268.84** |
| Carbon tax | Road | O–H-1-H-4-H-9-H-12-H-D | 588719.68 | 19.88 | 31984.48 |
| | Rail | O-R-1-R-4-R-9-R-12-R-D | 681533.84 | 18.32 | 10553.19 |
| | **Average values** | | **635126.76** | **19.100** | **21268.84** |
| Carbon trading | Road | O–H-1-H-4-H-9-H-12-H-D | 440353.82 | 19.88 | 31984.48 |
| | Rail | O-R-1-R-4-R-9-R-12-R-D | 580316.82 | 18.32 | 10553.19 |
| | **Average values** | | **510335.32** | **19.100** | **21268.84** |
| Carbon offset | Road | O–H-1-H-4-H-9-H-12-H-D | 440353.82 | 19.88 | 31984.48 |
| | Rail | O-R-1-R-4-R-9-R-12-R-D | 580316.82 | 18.32 | 10553.19 |
| | **Average values** | | **510335.32** | **19.100** | **21268.84** |

## 7. Conclusion

The mathematical models are developed for multimodal transportation planning problem with fuzzy demand and fuzzy time under different carbon emission policies, aiming to minimize both cost and time of low-carbon transportation. It is possible to provide more economical and environmentally friendly transportation selections for the multimodal transportation routing problem under different low-carbon policies by incorporating unpredictable factors and the impact of transport and transit capacities. Furthermore, the original SSA has been improved to solve the MOLCMTPP-FDFT by utilizing the number of iterations as a degree of freedom for $t$ distribution, which effectively balances local and global search capabilities of the SSA and compensates for its susceptibility to local optima. The experimental results demonstrate that mandatory carbon emission and carbon tax policies can successfully reduce carbon emissions, which provides appropriate approaches for urgent reduction in carbon emissions. The implementation of carbon trading policy can partially alleviate enterprise costs, thereby facilitating the early-stage promotion of low-carbon transportation practices.

ATDSSA has also been compared with other state-of-the-art intelligent optimization algorithms, such as GWO, PSO and NSGA-II. Although ATDSSA and NSGA-II obtain the same non-dominated solutions, the computation time required by ATDSSA is shorter than that of GSA, PSO and NSGA-II. Therefore, we can conclude that the proposed ATDSSA is a very competitive algorithm for solving multi-objective optimization problems. It can also be used to solve other multi-objective combinatorial optimization problems, such as multi-objective location routing problem, multi-objective vehicle routing problem, and multi-objective facility location problem. Additionally, the proposed mathematical model of MOLCMTPP-FDFT can obtain reasonable transportation routes while ensuring the minimization of cost, carbon emission and transportation time, which is helpful to the promotion of low-carbon transportation network.

The ATDSSA is a competitive algorithm that can effectively solve not only the multi-objective multimodal transportation planning problems, but also vehicle routing problem, location-routing problem and other combinatorial optimization problems. The mathematical model of MOLCMTPP-FDFT can provide cost-effective, environmentally-friendly and efficient transportation solution for logistics. However, this study also has two primary limitations. Firstly, due to the limited scope of this paper, a theoretical analysis of the convergence of ATDSS was not conducted. Secondly, there are many uncertainties in realistic transportation networks, but this study only considers fuzzy demand and time.

To extend this work and overcome its limitations, the following works can be done in the future:(1) exploring some effective methods to balance the global search and local search to further improve the performance of the algorithm. (2) proposing other methods to deal with uncertain variables, and reducing the impact of environmental changes on routing decision. (3) considering the low-carbon multimodal transportation of special products such as dangerous products, emergency supplies, and perishable food.

## CRediT authorship contribution statement

**Huizhen Zhang:** Conceptualization, Methodology, Software, Writing – original draft, Supervision, Writing – review & editing. **Qin Huang:** Validation, Formal analysis, Methodology, Software, Data curation, Writing – review & editing. **Liang Ma:** Writing – review & editing. **Ziying Zhang:** Investigation, Visualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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